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From Tweets to Ballots:
Quantifying and Correcting Structural Biases in
Social Media Polls through Post-Stratification

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Abstract

Election polls on social media are abundant, public, and timely, but the users who answer them are self-selected, so the crowd they capture is not representative of the voting population. The poll outcome reports the preference of a skewed audience rather than of the electorate. This thesis quantifies that distortion and corrects for it.

We treat a poll's framing as a measurable property of the instrument. The author of a poll makes several choices that may produce this distortion: the candidate order, the formality of how a candidate is named and the textual political leaning of the poll, sentiment and toxicity; we encode these framing choices and other cognitive biases with an election-agnostic pipeline as bounded markers. We fold these markers into a Sample-Calibrated Weighted Generalized Linear Model and we post-stratify its predictions onto the electorate. A zero-bias counterfactual projects the estimate under a neutral polling environment that strips the framing biases out while it preserves the genuine partisan and ideological composition of each stratum. The population weights correct who is in the sample; the counterfactual corrects how the poll is framed. We apply the same pipeline to the US 2016, 2020 and 2024 presidential elections.

The central finding is that the correction works in proportion to the representativeness the sample retains. It is near-exact when the audience still resembles the electorate, partial as the skew grows and weak when the audience collapses towards one pole. The framing markers improve the model in every cycle, but their leverage over the outcome erodes as the skew worsens, until framing no longer moves the result. The limit on the correction is the data source rather than the model: once a platform's user base decouples from the electorate, no amount of on-platform reweighting recovers the missing voters. The work contributes a reusable, election-agnostic pipeline and empirical evidence of how far the algorithmic debiasing of social media polls can go.

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Chapter 1

Introduction

Social media platforms let any user publish a poll, and during an election campaign they produce such polls in large numbers. On X, formerly Twitter, a poll is a tweet that carries a question, a short list of candidates and a public vote count. These polls are abundant, public and timely, and they cost almost nothing to gather, which makes them a tempting alternative to expensive phone surveys. They also carry one defining weakness. The users who answer them are self-selected: a person chooses whether to vote rather than being drawn into the sample at random. The crowd that results is not a probability sample of the electorate, so the headline figure of a social media poll measures the preference of a skewed audience rather than of the voting population.

A recent line of work collects these polls at scale and characterises their skew. Scarano et al. [17, 18], building on the Social Polls Project [21], assemble a large corpus of US presidential election polls from Twitter, document the biases the polls carry and link them to beliefs about voter fraud. We inherit this corpus and the problem it frames. Survey methodology offers a way to organise that problem. The total-error tradition, adapted to digital traces by the TED-ON framework [20], separates errors of representation, which concern who ends up in the sample, from errors of measurement, which concern what the instrument records. A social media poll can fail on both axes at once: the wrong people answer it, and the way it is phrased moves their answers.

Prior work addresses the two axes unevenly. It corrects the unrepresentative sample through post-stratification, the standard remedy that fits a model on the observed sample and reweights its predictions to the true composition of the population [7, 15, 24]. It treats the framing choices an author makes, the order of the options,

the way a candidate is named and the stance of the surrounding text, as biases to document and report rather than as quantities to fold into the correction. This leaves a gap. The framing of a poll is a measurable property of the instrument, yet it stays outside the model that produces the electoral estimate. Two questions follow, and this thesis takes them up: how much of the structural skew in social media polls can we measure and correct, and at what point does the correction break down?

We answer these questions with an election-agnostic pipeline that runs uniformly across the US 2016, 2020 and 2024 presidential elections. The pipeline encodes each framing choice as a bounded marker, the candidate order, the appellative formality and the textual political leaning of the poll, alongside sentiment, toxicity and a set of cognitive biases, so that the same logic measures every cycle and only the candidate names change. We fold these markers into a single-level Sample-Calibrated Weighted Generalized Linear Model (SCWG) and post-stratify its predictions onto the electorate. We do not use multilevel regression and post-stratification (MRP), the canonical instance of this idea; we keep a single-level weighted model and treat MRP as a yardstick. The novelty is twofold. First, we move the framing markers from the report into the model, as predictors of the recorded vote share. Second, we add a zero-bias counterfactual: we project the estimate under a neutral polling environment that strips the framing biases out while it preserves the genuine partisan and ideological composition of each stratum. The population weights correct representation; the counterfactual corrects measurement.

The work makes two contributions. The first is the pipeline itself: an election-agnostic, configuration-driven framework that extracts the framing markers, fits the SCWG and post-stratifies under the counterfactual, and that extends to a new contest by adding a registry entry rather than new code. The second is empirical. Across the three cycles the correction works in proportion to the representativeness the sample retains: it is near-exact when the audience still resembles the electorate, partial when the skew grows and weak when the audience collapses towards one pole. The markers improve the model in every cycle, but their leverage over the outcome erodes as the sample skew worsens, until framing no longer moves the result. The limit on the correction is the data source rather than the model.

This finding has a practical implication. A correction built on a platform’s own polls inherits the platform’s blind spots, and no amount of on-platform reweighting recovers voters that the platform has all but lost. Our results place a usable bound on how far the algorithmic debiasing of social media polls can go: the bound holds while the platform’s user base tracks the electorate and tightens sharply once the two decouple, as they do in the most skewed cycle. The thesis therefore reads as

much as a statement about the limits of the data source as about the method that corrects it.

The remainder of the thesis is organised as follows. Chapter 2 reviews the prior work that we build on and compare against. Chapter 3 formalises the data, the inferred controls and the population weights, and fixes the notation. Chapter 4 specifies the feature-extraction pipeline, the SCWG and the zero-bias counterfactual. Chapter 5 reports the correction across the three cycles. Chapter 6 interprets the results through the representation-versus-measurement split. Chapter 7 states the limitations of the design and the lines of work they open. Chapter 8 draws the takeaways together.

Chapter 2

Background and Related Work

This chapter reviews the work that this thesis builds on and compares against. We organise the review around five themes. First, we describe election polls on social media as a data source and the prior work that collected and characterised them (Section 2.1). Second, we present the total-error view that explains why the biases in such data matter (Section 2.2). Third, we survey the structural and linguistic biases that poll framing introduces (Section 2.3). Fourth, we cover the inference pipelines that supply our author- and audience-level controls (Section 2.4). Fifth, we discuss the platform forces that distort the sample and the post-stratification methods that correct for an unrepresentative one (Section 2.5).

2.1 Election Polls on Social Media

With a large portion of the population online and the high cost of phone-based surveys, querying people about their voter preference online can offer an affordable and timely alternative [10]. Social media platforms let any user publish a poll. On X (previously Twitter), a poll is a tweet that carries a question prompt, a short list of options, and a public vote count. The respondents to these polls are self-selected: a user chooses whether to vote rather than being drawn into the sample at random. Because participation is voluntary and open to anyone, the resulting set of voters is not a probability sample of the electorate.

The work we build on collected and characterised these polls at scale. Scarano et al. [17] assemble a large corpus of US presidential election polls from Twitter and study how to estimate candidate support from them. A follow-up by Scarano et al. [18] measures the prevalence of such polls, documents the biases they carry and

links them to voter-fraud beliefs. Both lines run through the Social Polls Project of Grabowicz et al. [21], which curates the per-cycle records that our dataset extends. We describe that extension in Chapter 3.

This prior work also marks the limits of the data. Scarano et al. [18] report a systematic gap between the public vote tally, which we observe, and a private count available only to poll authors, which excludes purchased or inauthentic votes. We treat this gap as an inherited property of the source. The same body of work frames the central problem: a social media poll reports the preferences of a skewed, self-selected crowd, not of the voting population. This thesis takes that problem as its starting point and asks how much of the skew we can measure and correct.

2.2 A Total-Error Perspective on Digital Traces

Treating a social media poll as a measurement of opinion invites a structured account of the points at which that measurement can go wrong, from who ends up in the sample to how the poll itself is phrased. Survey methodology offers one through the total survey error tradition [8], which separates errors of representation (who is in the sample) from errors of measurement (what the instrument records). Sen et al. [20] adapt this tradition to online behaviour with the TED-ON framework, a total-error framework for digital traces of human behaviour on online platforms.¹ The framework makes explicit that a digital trace is a proxy and that each step from the underlying behaviour to the recorded trace can introduce error.

This perspective organises the rest of the chapter and the rest of the thesis. The representation side covers who writes and answers a poll, which we address with author and audience controls (Section 2.4) and with post-stratification (Section 2.5). The measurement side covers how a poll is phrased and ordered, which we address with the bias markers that Section 2.3 motivates. Chapter 4 returns to this split and corrects for selection through population weights and for measurement through a zero-bias counterfactual.

2.3 Biases in Poll Framing

A poll author makes choices that can move votes independently of voter preference. Because these polls are routinely screenshotted, quoted and read as evidence of public opinion, such phrasing and ordering choices do not stay local: a single author's

¹We summarise the framework at the level we use it: a decomposition of error sources when a digital trace stands in for a behaviour. The exact set of error components is detailed in the original work.

framing can propagate into the wider signal that downstream readers treat as a measurement of opinion. We focus on three choices that recur across candidates and cycles and that carry established effects in the offline literature, which is what lets a single pipeline (Chapter 4) measure them uniformly rather than through cycle-specific rules. We group the relevant prior work by the choice it studies: option order, candidate naming and the stance of the surrounding text.

Option order. The order of options on a ballot or a poll can change the result. Miller and Krosnick [13] show that candidates listed first tend to receive a systematic vote gain that is independent of their actual support. Scarano et al. [17] observe the same positional effect in Twitter polls. This effect is well established offline and has been observed in Twitter polls, but prior work documents it rather than correcting an estimate for it; we encode it as the order bias marker that Chapter 4 defines and feed it into the correction.

Candidate naming. How a poll names a candidate also frames the reader. A formal title and a derogatory nickname carry different signals about the same person. Van den Berg et al. [22] study how names and titles frame political figures and propose a taxonomy of naming forms, from full formal titles down to pejorative pet names. We adopt this taxonomy directly as the basis of our formality bias marker. The taxonomy was built to analyse how figures are framed, not to estimate vote shares; we repurpose it as a predictor inside a model that produces an electoral estimate. The broader idea that word choice encodes a measurable stance toward another person is the subject of Pavalanathan et al. [16], who build a multidimensional lexicon for interpersonal stancetaking. Their work supports treating naming and affect in poll text as quantifiable rather than incidental.

Textual stance. Beyond names, the body text of a poll can lean toward one side. Detecting that lean reliably is difficult. Sen et al. [19] examine the reliability and validity of detecting approval of political actors in tweets and they show that such signals are noisy measures of true support, which prior work treats as a reason for caution rather than as something to fold into a correction. We take this caution seriously. We therefore use the textual lean as one control among several rather than as a direct vote signal and we keep it on the same bounded scale as the other markers (Chapter 4).

2.4 Inferring Author and Audience Attributes

Correcting for who produces a poll requires attributes that the raw poll does not state. Two inference pipelines supply these attributes and we take both as given rather than re-estimate them.

The first places users on a latent ideological scale. Barberá [1] estimates Bayesian ideal points from the political accounts a user follows, on the assumption that users follow accounts close to their own position. The output is a continuous partisan score per user. We use it for both the author and the audience of a poll.

The second infers demographics from a public profile. Wang et al. [25] present M3-Inference, a deep learning model that reads a profile name, screen name, image and description, and returns probability distributions over gender, age bracket and organisation status. We use its gender and age outputs as author-level controls. We keep gender as a probability rather than a hard label, because the inference is uncertain and because the gender of a political actor shapes how an audience responds, an effect that Voigt et al. [23] document in the RtGender corpus.

2.5 Platform Distortion and Sample Correction

The final theme is the gap between the platform sample and the electorate, and the methods that close it. The gap has two parts: the platform over-represents some users and the recorded scores are stretched relative to the general population.

Platform distortion. The platform does not show content at random. Gauthier et al. [6] document that the feed algorithm of X amplifies divisive and partisan content. The users and the scores we observe are therefore skewed toward engaged and ideologically extreme positions. We partly account for this in Chapter 3 by anchoring each population stratum to an in-sample quantile rather than to an absolute scale value. This anchoring addresses the platform skew within the sample only; it does not correct for it in full, and we defer that discussion to Chapter 7.

Correcting an unrepresentative sample. The standard remedy for a non-representative sample is post-stratification: fit a model on the observed sample, predict for each cell of a population grid and reweight the predictions by each cell's true share of the population. In political science the canonical instance is multilevel regression and post-stratification (MRP), which pairs a hierarchical model with a post-stratification grid [7, 15]. MRP corrects non-representative election polls in

practice, as Wang et al. [24] show when they recover accurate forecasts from a heavily skewed online sample.

Post-stratification also applies to social media polls in particular. Scarano et al. [17] post-stratify the same Twitter polls that we study and report election estimates with errors as low as one to two percentage points. Wang et al. [25] likewise pair demographic inference with post-stratification to build a more representative population sample. Post-stratification of social polls is therefore established prior art, and we treat it as the baseline that our work extends.

Our contribution is the integration of the framing-bias markers of Section 2.3 into the post-stratification model, together with a zero-bias counterfactual that we project across the population grid. The counterfactual strips out the framing biases while it keeps the genuine composition of each stratum. On the modelling side, we fit a single-level Sample-Calibrated Weighted Generalized Linear Model (SCWG) rather than a full hierarchical MRP; Chapter 4 specifies it. The combination corrects for selection through the population weights and for measurement through the counterfactual.

2.6 Summary

We have reviewed five strands of prior work and placed this thesis among them. Social media polls are an abundant but self-selected signal of political opinion, and prior work has both collected them at scale and documented that they misrepresent the electorate; this is the data and the problem we inherit. The total-error view gives us the scaffold: it separates errors of representation (who ends up in the sample), from errors of measurement (how the poll is framed), and the thesis corrects each side with a different tool. On the measurement side, the framing choices an author makes (option order, candidate naming and textual stance) have established effects, and we carry them forward as bounded markers rather than leaving them as phenomena merely to document. On the representation side, we lean on existing pipelines that infer the partisanship and demographics of who writes and answers a poll, and on post-stratification, the standard remedy for a non-representative sample. Our addition to that established practice is to fold the framing markers into the correction itself and to project the estimate under a neutral-framing counterfactual. Chapter 3 and Chapter 4 make these materials and methods precise.

Chapter 3

Data, Materials and Definitions

This chapter formalises the inputs to the analytical framework that we will further detail in Chapter 4. First, we explain how we build the base poll dataset from the raw social media records supplied by the Social Polls Project of Grabowicz et al. [21]. Second, we describe the author-level controls we obtain from external inference pipelines: partisan scores from Barberá’s Bayesian ideal point estimation [1] and demographic probabilities (gender and age bracket) from the M3-Inference model of Wang et al. [25]. Third, we describe the population weights that drive the post-stratification step, which we derive from official exit poll data. We close with the definitions and notation that Chapter 4 and 5 reuse. Throughout, we cover the three election cycles that the framework targets: the US 2016, 2020 and 2024 presidential elections.

3.1 Base Dataset Construction

3.1.1 Source

The raw data come from the Social Polls Project of SIMS Lab [21]. The data extend the poll collection studied by Scarano et al. [17, 18], sourced through X’s 10% sample (Decahose) and firehose APIs and largely following their standard JSON data schema.

The collection was built by querying the Twitter API for tweets containing the keywords "vote" or "voting" alongside the names of the major candidates in each cycle. This keyword-based sampling means polls that do not use these terms may be underrepresented in the dataset. Tweets containing polls were then manually vetted for relevance using a codebook that required each poll to: (i) refer explicitly to the

U.S. presidential election of the respective year, (ii) include both major candidates as options, (iii) ask about voting preference or election outcome prediction and (iv) be in English. Polls targeting a subset of voters, mentioning only one candidate, or using strongly one-sided framing were excluded as not relevant.

It is also worth noting that the public vote counts stored in the dataset may include purchased or otherwise inauthentic votes. Scarano et al. [18] document a systematic discrepancy between the public vote tally (which is what we observe) and a private count available only to poll authors, which excludes such votes. We treat this as an inherited limitation of the data source and discuss its implications in Chapter 7.

File role	Content	Key field
Polls	Tweet text, poll options, vote counts, author, meta-data	tweet id
Partisanship	User id $\rightarrow$ ideal point score	user id
Demographics	User id $\rightarrow$ M3-Inference probabilities	user id
Retweeters	Tweet id $\rightarrow$ list of retweeting users	tweet id
Favoriters	Tweet id $\rightarrow$ list of favouriting users	tweet id

Table 3.1. Raw input files per election cycle. Each cycle provides one or more poll files plus the four auxiliary files. File names and the number of source files vary by cycle; for example, US 2020 draws on three poll sources (`decahose`, `vote`, `voting`) and US 2024 draws on two (`general`, `harris`).

3.1.2 Poll record

Figure 3.1 shows a representative election poll as displayed on X. Each poll consists of a tweet body, which serves as the question prompt, and a set of options each carrying a label, a ballot position, and a vote count. Figure 3.2 illustrates the corresponding abstract record structure. We extract one record per poll-bearing tweet. A record stores the tweet identifier, the author identifier, the creation timestamp, the duration, the tweet text, the source label, the option list, the number of options n and the total vote count. The tweet identifier names the poll and we use the terms “poll” and “tweet” interchangeably at the record level.

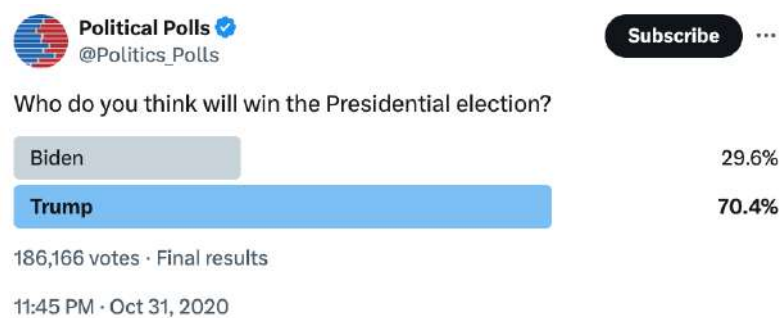


Figure 3.1. A poll as it appears on X (from Scarano et al. [18]). The key visible elements: the tweet text as the question prompt, the options with their labels and vote percentages and the total vote count.

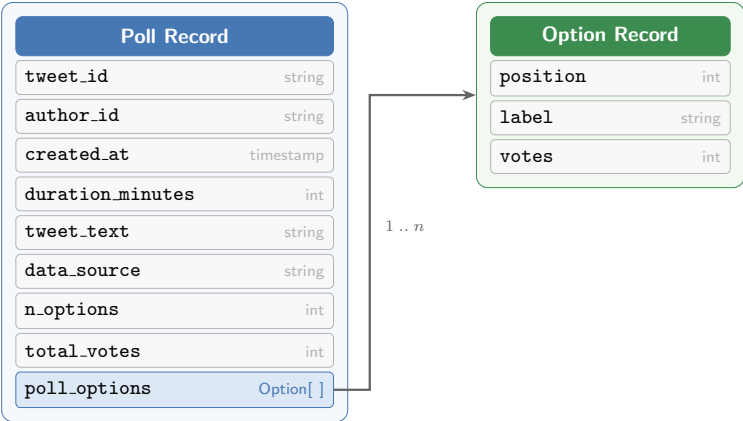


Figure 3.2. Abstract schema of a poll record. Each poll-bearing tweet yields one record with author and metadata fields, plus a list of n option records each carrying a ballot position, a label, and a vote count.

3.1.3 Filtering

We apply four filters before analysis.

- 1. **Zero-vote exclusion.** We drop polls with no recorded votes, since they carry no outcome signal.
- 2. **Deduplication.** When the same tweet identifier appears in more than one source file, we keep the record with the larger total vote count and discard the rest.
- 3. **Campaign-window cutoff.** For US 2024 we keep only polls created on or after 21 July 2024, the date on which the incumbent withdrew and the eventual

Democratic nominee entered the race. This aligns the poll window with the head-to-head contest that the cycle configuration targets. The other cycles use no date cutoff.

4. **Head-to-head restriction.** We retain a poll for modelling only when we match at least two configured candidates to its options. This restriction is enforced at the feature-extraction and modelling stages rather than at load time. It is the condition under which the outcome variable Y (Section 3.5) is defined: a poll without two matched candidates yields no positive- and negative-direction shares and leaves the analysis set.

We summarise the effect of these filters in Table 3.2.

Cycle	Raw polls	> 0 votes	Head-to-head
US 2016	778	761	758
US 2020	16336	7470	7427 ¹
US 2024	2786	2727	2712

Table 3.2. Dataset summary statistics per election cycle. “Raw polls” counts poll-bearing tweets before filtering. “> 0 votes” applies the zero-vote exclusion and deduplication. “Head-to-head” counts polls with at least two matched candidates, that is, the analysis set.

3.1.4 Missing values

The merge that joins polls to their inferred variables can leave gaps, for example, when an author has no partisan score in the inference pipeline. We describe how these missing values are handled at the modeling stage in Section 4.2.

3.2 Inferred User-Level Variables

The framework controls for who writes a poll and who engages with it. We obtain these user-level variables from external inference pipelines, as said in Section 3.1. We do not estimate them within this project. We describe the two pipelines below.

3.2.1 Partisan Scores

We use two partisan variables. The variable `author_partisanship` is the partisan score of the poll author. The variable `audience_mean_partisanship` is the mean partisan score of the users who engage with the poll. We compute both from a continuous score in which higher values indicate a more conservative position.

¹For US 2020, most polls come from the *decahose* source, which overlaps heavily with the other two and contains many polls captured immediately after publication, before they received any votes.

Estimation method

The scores come from the Bayesian ideal point estimation of Barberá [1]. The method places users on a latent liberal-conservative dimension from their following links to political accounts. It assumes that users follow accounts whose latent position lies close to their own. The model treats each follow as a function of the distance between the user position and the account position and it estimates the positions jointly by Bayesian inference. The output is a continuous score per user. We take this score as given from the inference pipeline.

Author score

The variable `author_partisanship` is the partisan score of the poll’s author, where higher values indicate a more conservative position.

Audience score

We define the audience of a poll as the set of distinct users who retweet or favourite it. For each such user with an available partisan score, we collect the score. The variable `audience_mean_partisanship` is the mean of these scores.

3.2.2 Demographic Inference

The framework controls for three author demographics. The variable `author_gender_male` is the probability that the author is male. The variables `author_age_30_39` and `author_age_40_over` are the probabilities that the author falls in the 30–39 bracket and the 40-or-older bracket. We combine M3’s two youngest brackets into a single under-29 group and treat it as the reference category, since it represents the most prevalent age group among Twitter users [17].

Estimation method

The probabilities come from the M3-Inference model [25]. M3-Inference is a deep learning model for demographic inference from social media profiles. It reads the name, the screen name, the profile image and the profile description and it returns probability distributions over gender, age bracket and organisation status. We take these probabilities as given from the inference pipeline and join them to each poll on the author identifier.

Gender inference and response

We use the gender probability as a control rather than as a label, since the inference is uncertain at the individual level. The gender of a political actor also shapes how an audience responds to a message, an effect documented in the RtGender corpus of Voigt et al. [23]. We therefore retain the gender probability as an author-level control, so that the model can separate the author demographic from the structural biases that Chapter 4 measures.

3.3 Population Weights for Post-Stratification

Post-stratification maps model estimates onto the electorate by reweighting predictions according to the true composition of the voting population and therefore requires the true share of each population stratum. We derive these shares from official exit poll data and we store one weight configuration per election cycle in a configuration registry. This design lets the framework extend to a new cycle by adding a registry entry.

3.3.1 Strata

We cross two partitions of the electorate. The partisan partition has three groups: Republican, Democrat and Independent. The ideological partition has three groups: Conservative, Moderate and Liberal. Unlike partisan affiliation, we do not infer ideological labels for individual users; no such inference pipeline is available in our data. Instead, the ideological dimension enters the post-stratification grid only at the population level, through exit poll shares, while its effect on the model is operationalized via the continuous text ideology score M_3 (Section 4.1.3), which is scaled by a fixed multiplier within each ideological stratum (Section 4.2.2).

The cross product yields nine strata. The weight of a stratum is the product of its partisan share and its ideological share. The weights sum to one over the nine strata, which matches the constraint $\sum_c W_c = 1.0$ later stated in Chapter 4.

3.3.2 Demographic profiles

Each partisan group carries a demographic profile: the male share and the age distribution of the group. We use the same exit poll source for these profiles as for the stratum weights.

3.3.3 Sources

For US 2020 we take the weights and profiles from the CNN national exit polls [3]. For US 2016 and US 2024 we use the corresponding national exit polls [2, 4]. Table 3.3 reports the US 2020 configuration as a worked example. The other cycles follow the same structure with different values.

Partisan group	Partisan share	Male share	Age 30–39	Age 40+
Republican	0.36	0.52	0.19	0.70
Democrat	0.37	0.43	0.27	0.49
Independent	0.27	0.48	0.23	0.60
Ideological shares: Conservative 0.38, Moderate 0.38, Liberal 0.24				

Table 3.3. US 2020 population weights and demographic profiles, from the CNN national exit polls [3]. The weight of a stratum is the product of its partisan share and its ideological share.

3.3.4 Mapping continuous predictors to strata

The partisan groups also fix the continuous predictors inside each stratum. We set the audience partisanship, the author partisanship and the text ideology predictor to a quantile of their observed distribution in the Twitter sample: the 75th percentile for Republicans, the 25th percentile for Democrats and the median for Independents. We use sample quantiles rather than absolute values, so that the frame stays robust to changes in how the partisan scale is calibrated across cycles. The ideological groups then scale the text ideology predictor by a fixed multiplier: 1.25 for Conservative, 1.00 for Moderate and 0.75 for Liberal.

3.4 Prediction Market Data

We complement the poll data with prediction market prices, which Chapter 5 uses as an external benchmark for our estimates. The prices come from PredictIt, a real-money political prediction market. We obtain the historical data directly from PredictIt under an academic data use agreement.

The data cover all three election cycles (US 2016, 2020 and 2024) at daily granularity. For each market contract and day, the record stores the open, high, low and closing prices together with the trading volume.

We use the daily closing price of each candidate’s contract. A contract price reflects the market’s implied win probability for that candidate, not a vote share [5, 26]. To

place the two major candidates on a common scale, we renormalise their two closing prices into a head-to-head probability,

$$\pi_{\text{pos}} = \frac{p_{\text{pos}}}{p_{\text{pos}} + p_{\text{neg}}}, \quad (3.1)$$

where p_{pos} and p_{neg} are the closing prices of the positive- and negative-direction contracts. The Results chapter compares π_{pos} against our post-stratified estimate and the official outcome.

3.5 Key Definitions and Notation

We close with the definitions that Chapter 4 and Chapter 5 will reuse. We fix each symbol here and keep it unchanged elsewhere.

3.5.1 Positive and negative direction

Each cycle defines one candidate as the *positive direction* of the bias scale and the opponent as the *negative direction*. We use the positive direction as the reference candidate for every directed quantity, such as the order bias M_1 , the formality bias M_2 and the outcome share. We list the convention in Table 3.4.

Cycle	Positive direction	Negative direction
US 2016	Trump	Clinton
US 2020	Trump	Biden
US 2024	Trump	Harris

Table 3.4. Positive- and negative-direction convention per election cycle. The positive direction is the reference candidate for every directed quantity.

3.5.2 Vote totals

For a poll i , we write N_i for the number of valid votes cast for the two major candidates, that is, the sum of the positive- and negative-direction vote counts. The raw record also stores an all-options vote count (Section 3.1.2), but the model restricts each contest to its two major candidates, so N_i excludes votes for any third option. N_i serves as the binomial variance weight in the model, where $W_i = N_i$.

3.5.3 Positive share

We write Y_i for the share of votes that the positive-direction candidate receives in poll i . We refer to this share as `positive_share` and define it as

$$Y_i = \frac{\text{positive-direction votes}}{\text{positive-direction votes} + \text{negative-direction votes}}, \quad Y_i \in [0, 1]. \quad (3.2)$$

The negative share is $1 - Y_i$, so the two shares sum to one by construction. Both require a head-to-head poll, since they need both candidates to be matched. The model uses Y_i as its outcome variable and weights each observation by the two-candidate total N_i .

3.5.4 Summary

This chapter has defined the three families of material that the framework consumes. We build the base poll dataset from the SIMS Lab records and four filters. We obtain partisan and demographic controls from external inference pipelines. We derive population weights from exit poll data and store them per cycle. We fix the direction convention, the vote total N_i and the outcome share Y_i . Chapter 4 uses these materials to extract the bias markers and to fit the post-stratified model.

Chapter 4

Methodology

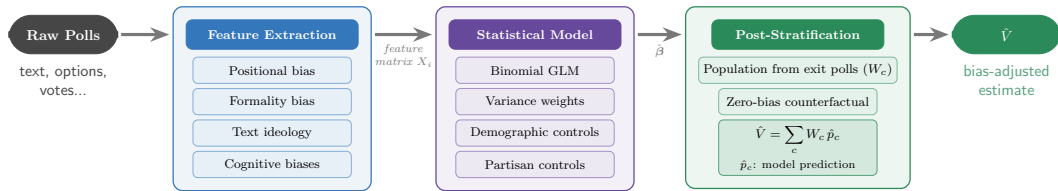


Figure 4.1. Overview of the three-phase methodology: bias features are extracted from raw polls, modelled via a sample-calibrated weighted GLM and projected onto the electorate through counterfactual post-stratification.

This chapter details the computational and statistical framework developed to quantify structural biases in social media polls and correct their electoral estimates. First, we leverage natural language processing (NLP) to encode theoretically-grounded forms of bias from polls contents into quantitative markers (4.1). These are then modeled statistically to project counterfactual estimates onto a representation of the voting population (4.2).

An overview of the complete pipeline is illustrated in Figure 4.2 at the end of this chapter. First, raw poll records (comprising poll text, structural voting choices and audience response metrics) are processed to extract structured bias metrics. These granular indicators are subsequently integrated into the statistical model to adjust the unrepresentative sampling inherent to social media environments.

4.1 Election-Agnostic Feature Extraction Pipeline

Comparing polls across election cycles is not straightforward. Each cycle involves different candidates, different political vocabularies, and different contextual references.

A poll from 2016 mentioning "Crooked Hillary" carries very different linguistic signals than one from 2024 mentioning "Sleepy Joe", yet both encode the same underlying phenomenon: informal, derogatory framing of a candidate. A naive approach would require writing separate, hand-crafted extraction rules for each cycle, which does not scale and makes cross-cycle comparisons unreliable, since the rules themselves would vary rather than the data.

The key insight is that while the surface forms differ across elections, the structural and linguistic mechanisms through which bias manifests remain consistent: candidates can always be placed first or last, named formally or pejoratively, and the surrounding text can always lean ideologically in one direction. If these mechanisms are extracted at a sufficient level of abstraction, parameterized by candidate names and election-specific vocabularies, but otherwise identical in logic, the same pipeline can operate uniformly across all cycles.

To achieve this, the pipeline extracts five main structural and linguistic features from each poll, defined in a way that is election-agnostic: the specific candidates and their known name variants are supplied through a configuration registry, while the extraction logic itself remains unchanged across the US 2016, 2020, and 2024 presidential elections.

4.1.1 Candidate Matching and Positional Order Bias

It has been shown in the political science literature that candidates listed first on a ballot or poll option list tend to receive a systematic vote inflation, independent of their actual support [13, 17].

Measuring this positional effect requires knowing which candidate occupies which slot in each poll. Since user-generated polls refer to candidates in highly varied ways, from full formal titles to nicknames, partial names, or emoji-decorated strings, a consistent matching step is necessary to reliably link each option to its canonical candidate. This is performed via an election-agnostic text normalization and candidate matching function. The function executes a multi-tiered filtering hierarchy:

1. **Exact Mapping:** Directly checks options against an inventory of known candidate variations compiled in the election configuration registry.
2. **Morphological Cleaning:** Removes candidate decorations via regular expressions, sifting through common instructional prefixes (e.g., *"Vote for"*, *"Support"*) and occupational or administrative titles (*"President"*, *"Senator"*, *"VP"*). It additionally trims election-specific temporal suffixes (e.g., *"2020"*, *"Administration"*, *"Campaign"*).

3. **Regex Fallback:** Applies regular expression matching with word-boundary constraints to process highly custom user inputs into canonical forms or categorize them as "Other".

Once options are identified, the pipeline tracks positional placement to capture the ordering effect bias (M_1).

For a defined target candidate (defined as the positive direction on our scale) and an opponent candidate (defined as the negative direction), the positional bias variable is formalized as:

$$M_1 = (Position_{\text{Opponent}} - Position_{\text{Target}}) \times 0.25 \quad (4.1)$$

where $Position \in \{1, 2, 3, 4\}$. The factor 0.25 is chosen so that the maximum possible value of M_1 (a four-option poll with the target candidate first and the opponent last, yielding $|M_1| = 0.75$) is commensurate in magnitude with $M_2 \in [-1, +1]$, keeping all bias markers on a comparable scale within the regression.

4.1.2 Formality and Appellative Taxonomy

The framing of political actors through names and titles alters audience perception. This framework evaluates candidate descriptions by implementing an automated linguistic classifier based on the naming formality taxonomy proposed by van den Berg et al. (2019)[22].

The taxonomy classifies mentions into a six-tier scale of decreasing formality, ranging from highly formal appellatives including titles, such as "President Donald Trump", down to first-name-only references like "Donald" or derogatory pet names like "Donnie":

Score	Category	Description	Linguistic Example
5	TFNLN	Title + First + Last Name	"President Donald Trump"
4	TLN	Title + Last Name	"Secretary Clinton"
3	FNLN	First + Last Name	"Kamala Harris"
2	LN	Last Name Only	"Biden"
1	FN	First Name Only	"Donald"
0	ADJ_PET	Negative Adjective / Pet Name	"Sleepy Joe", "Donnie"

Table 4.1. Formality Scale and Taxonomy Configurations.

The automated extraction pipeline handles text context across four specific steps:

1. **Named Entity Recognition (NER):** Spans are detected using a spaCy [14] English pipeline (`en_core_web_sm`) targeted at `PERSON` tags. The extraction window is expanded to include preceding tokens when a title or adjectival modifier is detected, ensuring that spans like "President Donald Trump" are captured whole rather than just the name itself.
2. **Named Entity Disambiguation (NED):** To eliminate false positives (e.g., distinguishing "*Hunter Biden*" from the candidate "*Joe Biden*"), candidate spans are validated through per-span API calls to a DBpedia Spotlight annotation endpoint, enforcing a strict confidence filtering threshold ($\tau = 0.50$). Spans resolving to non-candidate URIs are discarded.
3. **Span Deduplication:** Resolves multi-mention constraints within individual tweets by identifying overlapping boundaries and retaining the longest, most lexically informative span.
4. **Taxonomy Classification:** Evaluates the structural presence of components (Title, First Name, Last Name) using token-level regex match sequences.

To detect derogatory titles, the classification pipeline combines structural parsing with a valence audit utilizing a custom-weighted VADER (*Valence Aware Dictionary and sEntiment Reasoner*) [11] sentiment intensity engine augmented with candidate-specific derogatory adjectives drawn from the election configuration (e.g., *Sleepy* for Biden; *Orange* for Trump), each assigned a fixed negative valence score of -2.0 .¹ When a span contains a known derogatory adjective or manifests a significant drop in its compound sentiment score ($\Delta_{\text{sentiment}} \leq -0.05$) relative to the candidate’s baseline canonical name, it is assigned to the lowest tier (*Score* = 0).

The individual metrics computed for the target candidate (S_{Target}) and the opponent (S_{Opponent}) are transformed into a symmetric continuous formality bias metric (M_2) bounded inside $[-1.0, +1.0]$:

$$M_2 = \frac{S_{\text{Target}} - S_{\text{Opponent}}}{6.0} \quad (4.2)$$

4.1.3 Textual Political Leaning via Zero-Shot NLI

Beyond overt nominal framing, the body text of a poll tweet often carries partisan sentiment or ideological presuppositions. To extract this latent ideological align-

¹This augmentation is applied only within the formality classifier; the sentiment and toxicity extraction step uses the unmodified VADER lexicon to preserve general-purpose sentiment calibration.

ment without relying on hand-labeled training samples, we frame political leaning extraction as a zero-shot Natural Language Inference (NLI) task.

The framework processes tweet bodies using a large-scale transformer architecture, specifically `facebook/bart-large-mnli` [12]. For each poll text, the framework evaluates a set of six election-specific hypotheses, covering ideological orientation (*conservative* and *liberal* politics) and direct candidate attitudes (praise or criticism of each candidate):

- H_{pos} : *"The text supports conservative politics"*
- H_{neg} : *"The text supports liberal politics"*
- H_{praise} : *"The text praises [Target/Opponent Candidate]"*
- $H_{\text{criticize}}$: *"The text criticizes [Target/Opponent Candidate]"*

For each hypothesis H , the model returns logits across three classes: entailment, neutral, contradiction. To isolate the directional signal from neutral noise, we normalize entailment against contradiction:

$$\text{score}(H) = \frac{P(\text{entailment})}{P(\text{entailment}) + P(\text{contradiction})} \quad (4.3)$$

All six scores enter the analysis. We designate $M_3 = \text{score}(H_{\text{pos}})$, the entailment score for the conservative-politics hypothesis, as the headline ideological marker, since it is the most direct continuous proxy for the poll’s slant toward the positive-direction candidate. This bounds $M_3 \in [0, 1]$, where values approaching 1.0 signify a clear structural alignment with the target candidate’s ideological domain. The remaining five scores (the liberal-politics score $\text{score}(H_{\text{neg}})$ and the four candidate-directed praise and criticism scores) are retained as auxiliary leaning predictors and enter the regression alongside M_3 ; the same six scores are reused in the correlation analysis of Chapter 5.

4.1.4 Sentiment and Toxicity Extraction

Linguistic features are complemented by extracting undirected sentiment and text toxicity scores. The overall emotional intensity of a poll is captured by computing the compound valence score through standard VADER [11] sentiment analysis alongside a sentiment intensity feature defined as $1.0 - \text{Neutrality}$.

To capture aggressive linguistic environments which strongly induce expression bias, we detect toxicity using `Detoxify` [9]. Specifically, we utilize the multi-label `unbiased` model, which has been explicitly optimized to reduce false positive flags triggered by the simple presence of politically sensitive or demographic identity terms. This model maps the combined text fields directly to a continuous toxicity metric: $T_s \in [0, 1]$.

4.1.5 Cognitive Bias Extraction via LLM

Polls often embed subtle cognitive biases, such as leading questions or social desirability pressure. Traditional NLP techniques struggle to capture these abstract psychometric properties. We employ a Large Language Model (`gemini-3.1-flash-lite`) to evaluate the semantic relationship between the poll prompt and its options.

We instruct the LLM to act as a cognitive scientist and identify the presence of six specific biases: Confirmation Bias, Anchoring, Availability Heuristic, Social Desirability Bias, Acquiescence Bias and Demand Characteristics Bias. To ensure reproducibility and strict structural outputs, we supply the model with the system prompt detailed in Appendix B.1.

4.2 Sample-Calibrated Weighted Modeling and Post-Stratification

The primary contribution of our methodology (extending the Social Polls framework of Grabowicz et al. [21]) is the integration of these extracted bias markers into a frequentist framework designed to weight and adjust unrepresentative samples: the SCWG.

4.2.1 Generalized Linear Model Specification

Let i denote an individual social media poll, and let N_i denote its number of valid votes for the two major candidates. Let Y_i represent the share of those votes cast in favor of our designated positive-direction target candidate ($Y_i \in [0, 1]$). Because the outcome variable represents a count of successes over a finite sequence of independent user trials, we model Y_i using a Generalized Linear Model specification with a Binomial error distribution family and a Logit link function.

Missing values in the bias predictors inherited from the merge step (Section 3.1.4) are imputed with the neutral value 0. Polls that lack a valid outcome Y_i are dropped from the analysis set.

The complete mathematical specification of the linear predictor η_i is defined as follows:

$$\begin{aligned} \eta_i = \ln\left(\frac{\mu_i}{1 - \mu_i}\right) = & \beta_0 + \beta_1 \cdot \text{AudiencePartisanship}_i + \beta_2 \cdot \text{AuthorPartisanship}_i \\ & + \beta_3 M_{1,i} + \beta_4 M_{2,i} + \beta_5 M_{3,i} + \sum_j \lambda_j \cdot \text{NLI}_{j,i} \\ & + \beta_6 \text{Sentiment}_i + \beta_7 \text{Intensity}_i + \beta_8 \text{Toxicity}_i \\ & + \beta_9 \ln(1 + \text{Duration}_i) \\ & + \sum_k \gamma_k \cdot \text{CognitiveBias}_{k,i} + X_i \delta \end{aligned} \quad (4.4)$$

where $\mu_i = \mathbb{E}(Y_i)$, $M_{1,i}$ is the candidate order bias, $M_{2,i}$ is the formality bias, and $M_{3,i}$ is the zero-shot text ideology metric. The vector $\text{CognitiveBias}_{k,i}$ incorporates binary indicators identifying distinct heuristic biases (e.g., confirmation, anchoring, availability, social desirability) detected within the prompt framing. The matrix $\mathbf{X}_i$, as detailed in Chapter 3 (Data), stores author-level demographic and partisan controls inferred through external inference pipelines (e.g., M3-Inference model [25] and Bayesian ideal point estimation) [1]. The term $\sum_j \lambda_j \cdot \text{NLI}_{j,i}$ collects the five auxiliary zero-shot leaning scores of Section 4.1.3: the liberal-politics score and the four candidate-directed praise and criticism scores.

Crucially, to account for heteroskedasticity and scale the structural variance across vastly different poll sizes, the optimization function weights each observation by its total trial size. The variance weights W_i applied during maximum likelihood estimation are equivalent to the absolute counts of poll participants:

$$W_i = N_i \quad (4.5)$$

This forces the estimation process to prioritize high-density observations, mirroring traditional survey practices where larger sample sizes yield a tighter variance profile.

We additionally include poll duration as a structural control. Short polls capture impulsive, highly partisan early responders, whereas multi-day polls accumulate a broader and more representative audience. We enter it as $\ln(1 + \text{Duration}_i)$ to compress its long right tail, and hold it at the global median in the post-stratification frame.

4.2.2 Target Population Stratification

To map these model estimates back onto the actual electorate, we construct a joint post-stratification grid crossing political, ideological and demographic variables. For

a chosen historical election cycle, the true voting population is partitioned into distinct mutually exclusive strata (cells) indexed by c . The total population weight of a specific stratum, W_c , is derived from official national exit poll databases and census records [3], satisfying:

$$\sum_c W_c = 1.0 \quad (4.6)$$

The population is partitioned into nine strata by taking the Cartesian product of two independent taxonomies drawn from exit poll data: *Partisan Affiliation* (Republican, Democrat, Independent), reflecting party identification, and *Ideological Orientation* (Conservative, Moderate, Liberal), reflecting self-reported ideology. Each stratum’s weight is the product of its partisan share and its ideological share.

For each partisan stratum, the continuous predictors `audience_mean_partisanship`, `author_partisanship`, and `positive_ideology_score` are set to the corresponding quantile of their observed distributions in the Twitter sample: the 75th percentile for Republicans, the 25th percentile for Democrats, and the median (50th percentile) for Independents. Using sample quantiles rather than absolute scale values makes the frame robust to shifts in how partisanship scores are calibrated across election cycles.

The liberal-politics leaning score is set symmetrically, to the inverted quantile $1 - q$ of its in-sample distribution, where q is the quantile used for the conservative score in the same stratum (the Republican stratum takes the 25th percentile, the Democrat stratum the 75th, and the Independent stratum the median). Treating the two anti-correlated leaning scores symmetrically keeps each stratum inside the region of feature space the model was trained on: zeroing the liberal score while letting the conservative score vary would feed the Democrat stratum an out-of-distribution pairing (low conservative leaning with zero liberal leaning) that the training data rarely contains.

Demographic controls are set to the known gender and age distributions of each partisan group, drawn from the same exit poll data used to derive the stratum weights [3].

4.2.3 Why in-sample quantiles rather than population values

The platform sample is not a random draw from the electorate. The feed algorithm of the platform amplifies divisive and partisan content, an effect that Gauthier et al. [6] document for the X (previously Twitter) feed. The partisan scores that we observe are therefore stretched relative to the general population, since the platform

over-represents engaged and ideologically extreme users. Two consequences follow for the post-stratification frame.

First, the model is fit on this skewed sample. If we set the stratum predictors to absolute population or census values, we ask the model to predict at input values that lie outside the range it has seen. The fitted surface then extrapolates and the post-stratified estimate becomes unstable. Anchoring each stratum to a quantile of the observed in-sample distribution keeps every stratum inside the support of the training data, so the model interpolates rather than extrapolates.

Second, the quantile choice scopes the correction. The stratum weights W_c carry the real-world composition of the electorate, since they come from exit poll shares. The predictor values stay in-sample, so they carry the relative position of each partisan group within the platform. The frame thus corrects for within-platform sample skew through the predictors and for population representativeness through the weights. We do not map the platform scale onto an absolute real-world scale, because the feed-driven distortion makes that mapping unreliable.

4.2.4 Zero-Bias Counterfactual Post-Stratification

Once the SCWG parameters $(\hat{\beta}, \hat{\gamma}, \hat{\delta})$ are estimated on the data, we project a perfectly neutral polling environment across the target population strata. This is accomplished by generating a synthetic post-stratification frame where all artificial structural and linguistic bias markers are strictly minimized or set to zero.

Specifically, inside this prediction matrix, we enforce a zero-bias counterfactual constraint:

$$\begin{aligned} M_{1,c}^* &= 0, & M_{2,c}^* &= 0, & M_{3,c}^* &= \text{Baseline}_c^{\text{pos}}, & \text{NLI}_c^{\text{neg}} &= \text{Baseline}_c^{\text{neg}}, \\ \text{NLI}_{j,c}^{\text{praise/crit}} &= 0, & \text{Sentiment}_c^* &= 0, & \text{Intensity}_c^* &= 0, \\ \text{Toxicity}_c^* &= 0, & \text{CognitiveBias}_{k,c}^* &= 0, & \text{Duration}_c^* &= \text{median} \end{aligned} \quad (4.7)$$

We retain the genuine ideological composition of each stratum on both leaning axes: the conservative marker M_3 keeps its stratum baseline $\text{Baseline}_c^{\text{pos}}$ and the liberal marker keeps its symmetric baseline $\text{Baseline}_c^{\text{neg}}$. The four candidate-directed praise and criticism scores are creator-induced framing signals and are zeroed with the other structural markers. Poll duration is a structural property of the genuine polling environment rather than a framing choice, so it is held at the sample median rather than zeroed.

This counterfactual setting strips away the structural distortions introduced by the poll creator’s phrasing, item ordering, and platform toxicity, while preserving the natural demographic controls ($\mathbf{X}_c$) and baseline partisan characteristics unique to stratum c .²

The model generates an adjusted probability prediction for each population stratum using the inverse link function:

$$\hat{p}_c = \frac{1}{1 + \exp(-\hat{\eta}_c^*)} \quad (4.8)$$

The final, post-stratified adjusted election estimate ($\hat{V}$) for the target candidate is computed by aggregating these strata predictions weighted by each stratum’s share of the actual electorate, as measured by national exit polls:

$$\hat{V} = \sum_c W_c \cdot \hat{p}_c \quad (4.9)$$

This integration corrects for both selection bias (by reweighting underrepresented demographic groups through W_c) and measurement bias (by evaluating predictions under a simulated zero-bias counterfactual configuration $\hat{\eta}_c^*$).

4.3 Implementation Details

The pipeline is implemented in Python, organized as a **Typer** command-line application. Each feature extraction step is exposed as a standalone sub-command, which allows individual steps to be run independently or in sequence via the top-level `run_pipeline.py` runner. The runner also accepts flags to skip any phase, which is useful when rerunning only the modeling step after a configuration change.

4.3.1 Libraries

Named entity recognition and span expansion use spaCy [14] with the `en_core_web_sm` pipeline. Political leaning classification uses the Hugging Face **transformers** library to load and run `facebook/bart-large-mnli` [12]; on CUDA-capable hardware, the model is loaded in FP16 (Half-Precision Floating Point) precision, reducing inference

²The rationale for the asymmetry is the two ideology markers (M_3 and the liberal score) reflects the genuine ideological composition of voters within each stratum, not a distortion introduced by the poll creator. Zeroing them would amount to assuming all population strata share the same baseline political orientation, which would invalidate the post-stratification logic. The zero-bias constraint therefore targets only *creator-induced* structural biases, while preserving each stratum’s authentic partisan and ideological identity.

time from approximately two hours to under twenty minutes on the full dataset. Sentiment analysis uses the `vaderSentiment` [11] library and toxicity detection uses the `detoxify` [9] package. The GLM and OLS models are fitted with `statsmodels`. LLM-based cognitive bias detection calls the Google GenAI API.

4.3.2 Data formats

Each feature extraction step reads from the base dataset (JSONL or CSV) and writes one output CSV per election to `data/processed/<election>/`. The cognitive bias step writes JSONL, with one record per poll. The modeling step reads and merges all processed CSV files before fitting the GLM.

4.3.3 Time-series dashboard

In addition to the single global estimate, the visualization phase applies the fitted SCWG model to rolling 7-day windows over the campaign period. For each window containing at least three polls, a stratum-specific post-stratification frame is built and the weighted average is computed, producing a daily time series of bias-adjusted vote-share estimates.

Within each window we do not recompute the partisan quantile anchors, which would be unstable given the small number of polls per window. Instead we freeze the anchors to the full campaign sample and shift them by the window’s mean deviation from the global mean. This keeps the relative partisan ordering of the strata fixed (Republican > Independent > Democrat) while letting genuine week-to-week movement in the absolute level propagate into the post-stratified estimate.

This series, together with the raw average, the baseline estimate, and campaign milestone markers (debates, conventions), is exported as a self-contained HTML dashboard.

Appendix A.1 shows the core GLM fitting and post-stratification functions.

4.4 Architectural Synthesis

To consolidate the methodologies detailed in this chapter, Figure 4.2 provides a comprehensive visual summary of the complete analytical pipeline.

The architecture is explicitly designed to translate unstructured digital traces into statistically sound electoral estimates. As illustrated, the pipeline operates in three sequential phases:

1. **Feature Extraction:** Raw poll data is passed through parallel NLP routines. These routines extract bounded, continuous bias markers from raw polls representing positional advantage (M_1), naming formality (M_2), textual ideology (M_3), toxicity and heuristic cognitive biases.
2. **Statistical Modeling:** The extracted metrics, alongside demographic controls, are aggregated into an integrated feature matrix ($\mathbf{X}_i$). This matrix feeds the Sample-Calibrated Weighted GLM, which weights the variance of each observation by its total trial size ($W_i = N_i$) to prevent noisy, low-engagement polls from skewing the parameters.
3. **Post-Stratification:** The fitted model is applied to a target population grid. By enforcing a zero-bias counterfactual constraint across all strata, the pipeline neutralizes the measured structural distortions, ultimately aggregating the strata predictions into a single, bias-adjusted election estimate ($\hat{V}$).

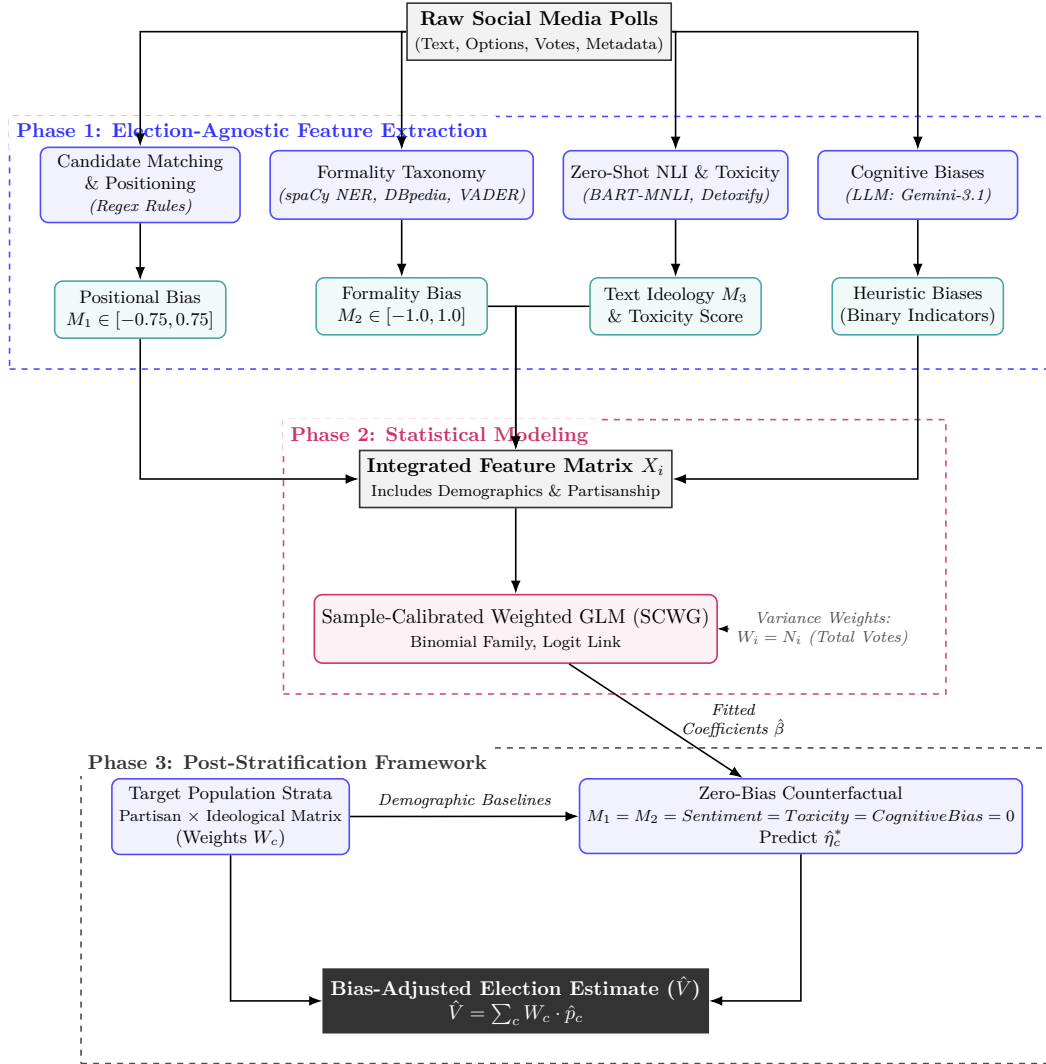


Figure 4.2. Architectural overview of the pipeline: from raw social media polls to bias-adjusted electoral estimates via feature extraction, SCWG estimation, and counterfactual post-stratification.

Chapter 5

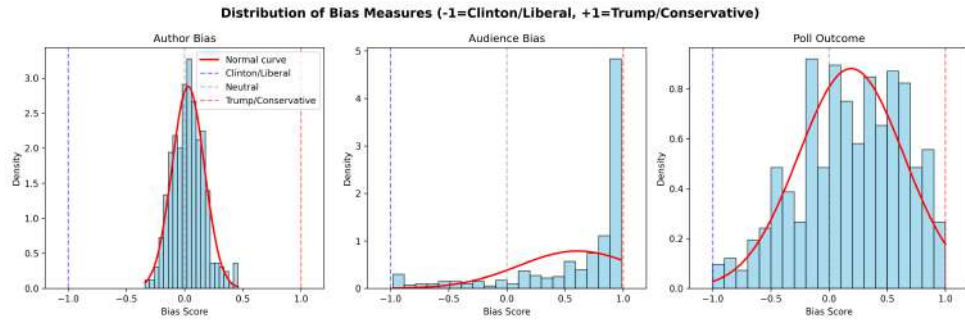
Results

This chapter reports the outcome of applying the framework of Chapter 4 to the three election cycles introduced in Chapter 3: the US 2016, 2020 and 2024 presidential elections. We organise the results around the questions the framework is built to answer. First, we describe the sample the polls draw from and show how unrepresentative it is (Section 5.1). Second, we measure the marginal effect of each structural bias marker on the recorded vote share (Section 5.2). Third, we report the correlation structure that links the markers to author bias, audience bias and poll outcome (Section 5.3). Fourth, we ask whether the markers improve the statistical model (Section 5.4). Fifth, we report the post-stratified estimate against the official result (Section 5.5). Finally, we track the estimate over the campaign and benchmark it against a prediction market (Section 5.6).

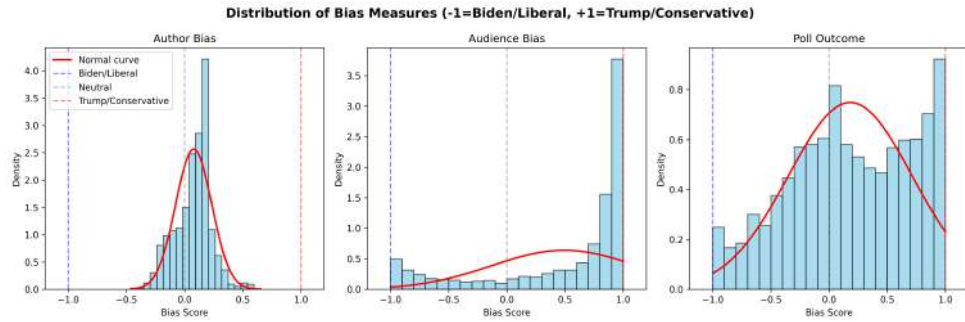
One result recurs through the chapter and we state it at the outset. The framework corrects the raw poll signal towards the truth in every cycle, but the size of the residual bias grows with the skew of the underlying sample: the correction is near-exact in 2020, partial in 2016 and weakest in 2024.

5.1 The Representativeness of the Poll Sample

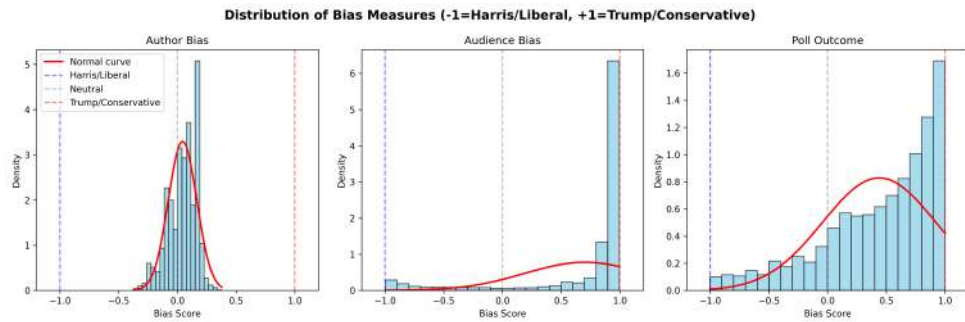
The framework exists because the poll sample is not the electorate. We make this concrete before reporting any correction. For each cycle we examine the distribution of three quantities on the bias scale of Chapter 3, which runs from -1 (the negative-direction candidate, liberal pole) to $+1$ (the positive-direction candidate, conservative pole): the author bias, the audience bias and the poll outcome. Figure 5.1 shows the three distributions for each cycle.



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.1. Distribution of author bias, audience bias and poll outcome on the $[-1, +1]$ bias scale, with a fitted normal curve. The author distribution is approximately normal and slightly right of neutral in every cycle. The audience distribution is heavily right-skewed and the skew worsens across cycles, concentrating at the +1 boundary in 2024.

The author distribution is the most regular of the three. In every cycle it is approximately normal, centred slightly right of neutral, with most mass between -0.5 and $+0.5$ and a peak near $+0.1$. Poll authors lean marginally towards the conservative pole on average, and the 2024 authors form a tighter, more homogeneous group than those of 2016 and 2020.

The audience distribution drives the central problem. In all three cycles it is heavily right-skewed: little mass lies below the conservative pole and density rises sharply towards the $+1$ boundary. The skew worsens monotonically across cycles. In 2016 the mass concentrates near $+1$; in 2020 the distribution is sharply bimodal with a spike at $+1$; in 2024 it is almost flat from -1 to $+0.9$ and then spikes at $+1$, the most extreme of the three. The audience that votes in these polls is therefore overwhelmingly conservative and increasingly so. This single feature explains the raw overestimation of Trump support and, as Section 5.5 shows, the limit on how much any reweighting can recover. The 2020 and 2024 cycles also straddle a change in the platform itself, from Twitter to X; we examine its role in Chapter 6.

The poll outcome distribution is right-skewed in every cycle, with more Trump-winning polls than opponent-winning ones. The skew is mildest in 2016 and sharpest in 2024, where a tall spike at $+1$ marks the many polls that record a 100% Trump result.

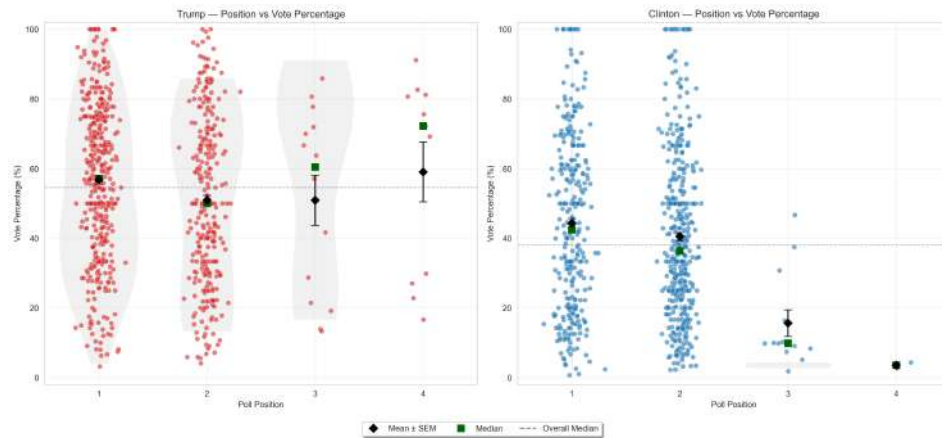
5.2 Marginal Effects of the Structural Bias Markers

We now measure how each structural framing choice moves the recorded vote share. For each marker we report the vote share of the reference candidate across the levels of that marker. We treat these as descriptive marginal effects: they combine the genuine preference of the audience with the framing effect of the marker and Section 5.4 reports the model that separates the two. The markers are the candidate order (M_1), the appellative formality (M_2) and the textual political leaning (M_3) of Chapter 4.

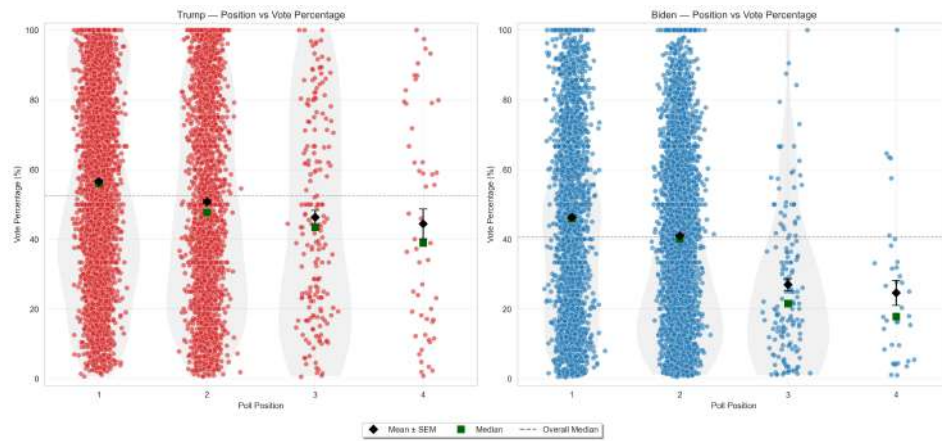
5.2.1 Candidate Order

The candidate listed first on a poll tends to gain votes independently of support [13, 17]. Figure 5.2 shows the vote share of each candidate by ballot position and Tables 5.1–5.3 give the per-position figures.

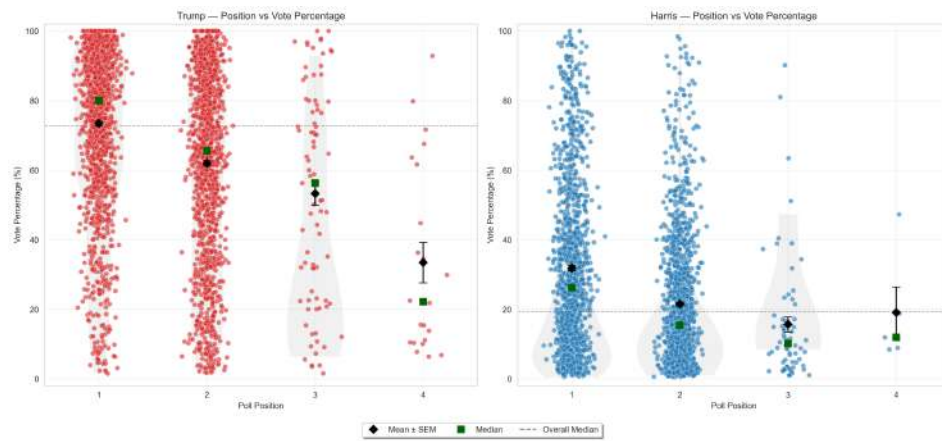
A primacy effect appears in every cycle and for both candidates. In 2016 Trump falls from 56.7% at position 1 to 50.9% at position 2 and Clinton falls from 44.2% at



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.2. Vote percentage by ballot position for each candidate. Grey violins show the per-position distribution; black diamonds mark the mean $\pm$ SEM, green squares the median and the dashed line the overall median. A primacy effect is visible in every cycle: both candidates lose share as their position moves from first to last.

position 1 to 40.5% at position 2, with sharp drops at the rare positions 3 and 4. In 2020 the effect is monotonic across all four positions: Trump moves 56.6% $\rightarrow$ 50.7% $\rightarrow$ 46.2% $\rightarrow$ 44.4% and Biden 46.3% $\rightarrow$ 40.9% $\rightarrow$ 27.0% $\rightarrow$ 24.6%. In 2024 the primacy effect is the strongest of the three cycles: Trump moves 73.4% $\rightarrow$ 61.9% $\rightarrow$ 53.3% $\rightarrow$ 33.4% and Harris 31.8% $\rightarrow$ 21.5% $\rightarrow$ 15.6%. The reference candidate occupies the first slot in the majority of polls in every cycle, so the primacy effect adds to the overestimation rather than cancelling out.

Position	Trump		Clinton	
	<i>N</i>	Mean %	<i>N</i>	Mean %
1	414	56.7	304	44.2
2	285	50.9	411	40.5
3	14	50.9	13	15.6
4	11	59.0	2	3.5
Overall	724	54.3	730	41.5

Table 5.1. US 2016 vote percentage by ballot position.

Position	Trump		Biden	
	<i>N</i>	Mean %	<i>N</i>	Mean %
1	4,237	56.6	2,605	46.3
2	2,449	50.7	4,121	40.9
3	190	46.2	157	27.0
4	53	44.4	38	24.6
Overall	6,929	54.1	6,921	42.5

Table 5.2. US 2020 vote percentage by ballot position.

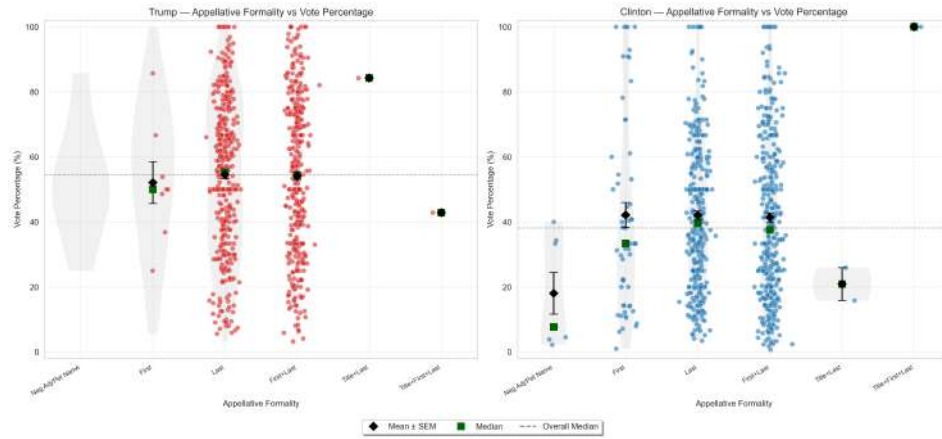
Position	Trump		Harris	
	<i>N</i>	Mean %	<i>N</i>	Mean %
1	1,323	73.4	1,215	31.8
2	1,232	61.9	1,266	21.5
3	83	53.3	68	15.6
4	22	33.4	5	19.0
Overall	2,660	67.1	2,554	26.2

Table 5.3. US 2024 vote percentage by ballot position.

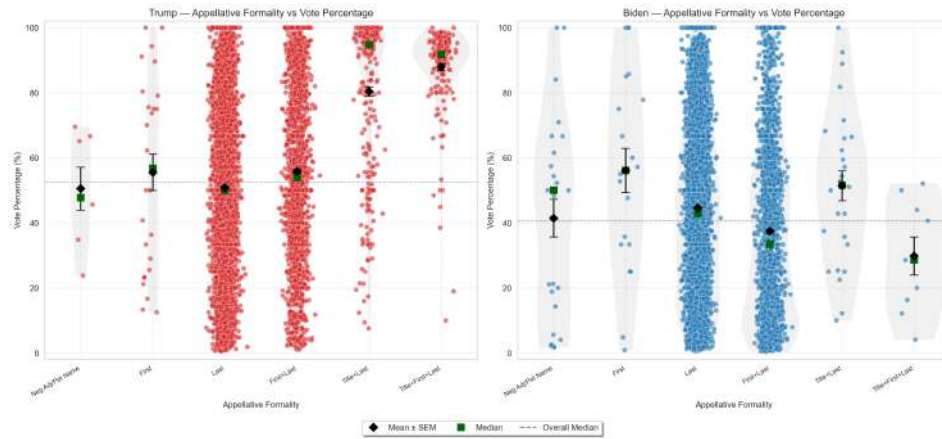
5.2.2 Appellative Formality

The way a poll names a candidate frames the reader [22]. Figure 5.3 shows the vote share by appellative category, from a negative adjective or pet name (the lowest formality) up to a title with first and last name (the highest).

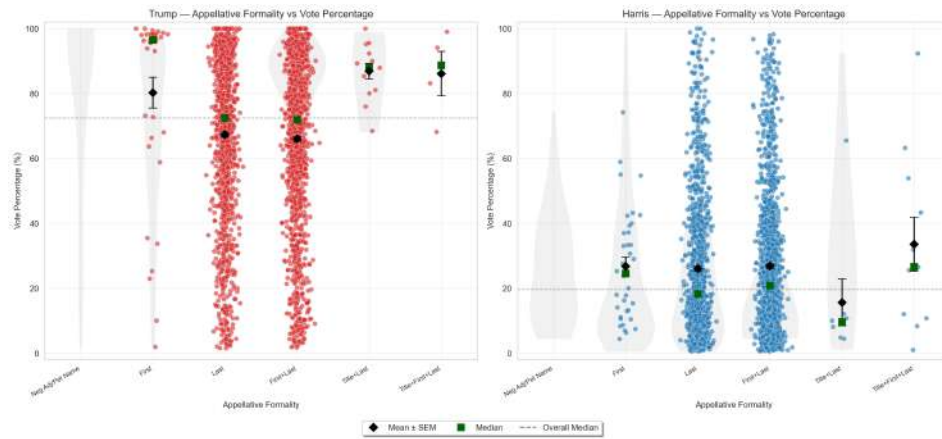
The formality effect is uneven across candidates and cycles. In 2016 Trump’s share is stable near 54–55% across the common categories, while Clinton’s falls steeply under derogatory naming: the seven polls that use a pet name (for example, “Crooked Hillary”) return a mean of only 18.0% for her, far below her overall median. In 2020 the striking pattern is the inflation of Trump’s share under high-formality titles: “President Trump” (title and last name) returns a mean of 80.3% and “President Donald Trump” (title, first and last name) returns 87.8%, against an overall mean of 54.1%. Biden shows no comparable gain and his most formal mention even drops to 29.8%. In 2024 the effect disappears: no derogatory naming occurs for either candidate, Trump’s share is uniformly high (roughly 66–67% across the common categories, even higher under rare ones) and Harris’s is uniformly low (roughly 26%). By 2024 the way a candidate is named no longer discriminates between high and low vote shares, because the structural skew dominates the outcome regardless of phrasing.



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.3. Vote percentage by appellative formality category for each candidate. The effect of formality is candidate- and cycle-specific: derogatory naming depresses Clinton’s share in 2016, high-formality titles inflate Trump’s share in 2020 and formality has no clear effect in 2024.

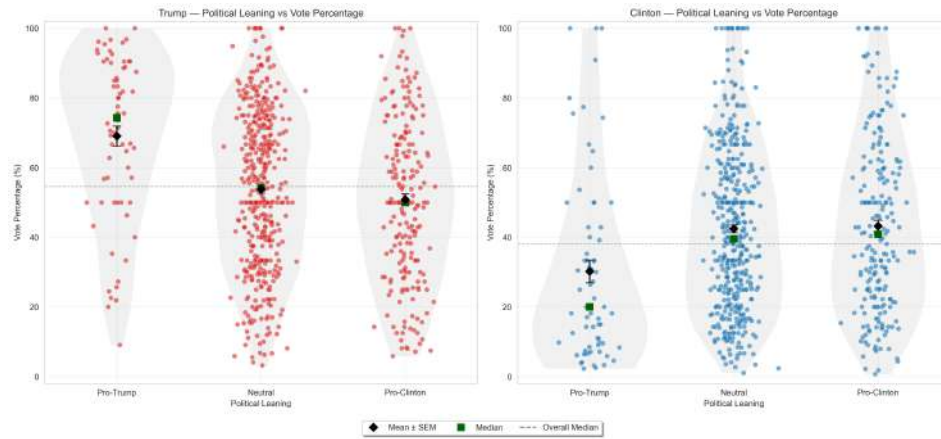
5.2.3 Textual Political Leaning

Beyond the candidate names, the body text of a poll can lean towards one side. Figure 5.4 shows the vote share by the political leaning of the poll text (pro-Trump, neutral, or pro-opponent) and Table 5.4 gives the reference-candidate means.

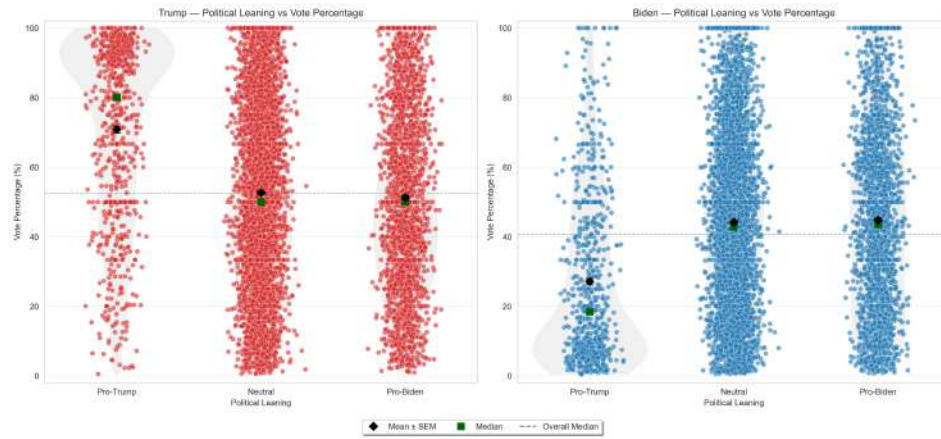
Text leaning	US 2016	US 2020	US 2024
Pro-Trump	69.0	70.8	66.4
Neutral	53.8	52.6	67.0
Pro-opponent	50.8	51.2	67.4

Table 5.4. Mean Trump vote percentage by political leaning of the poll text. The pro-Trump premium of 2016 and 2020 vanishes in 2024, where the three categories are statistically indistinguishable.

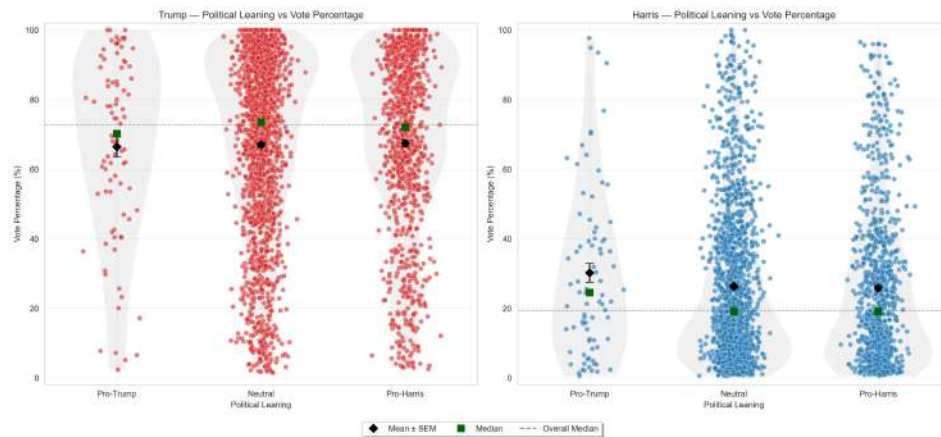
In every cycle the poll texts skew towards the opponent: roughly 60% are neutral, with pro-opponent texts outnumbering pro-Trump texts by about three to one in 2016 and 2020 and by about ten to one in 2024. In 2016 and 2020 the leaning still moves the outcome in the expected direction: pro-Trump text raises Trump’s mean share to 69.0% (2016) and 70.8% (2020), well above the neutral level near 53%. The most striking single result of the chapter appears in 2024. There Trump’s mean share is essentially identical across the three leaning categories: 66.4% under pro-Trump text, 67.0% under neutral text and 67.4% under pro-Harris text. The political leaning of the text carries no signal for the outcome in 2024 because the audience composition overwhelms it.



(a) US 2016



(b) US 2020



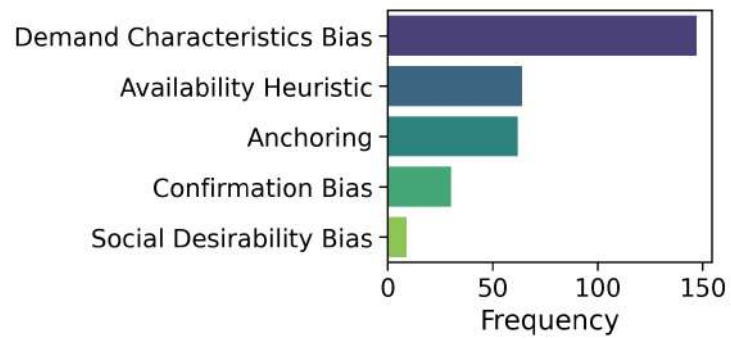
(c) US 2024

Figure 5.4. Vote percentage by the political leaning of the poll text. In 2016 and 2020 pro-Trump text raises Trump's share well above the neutral level; in 2024 Trump's share is flat across all three leaning categories.

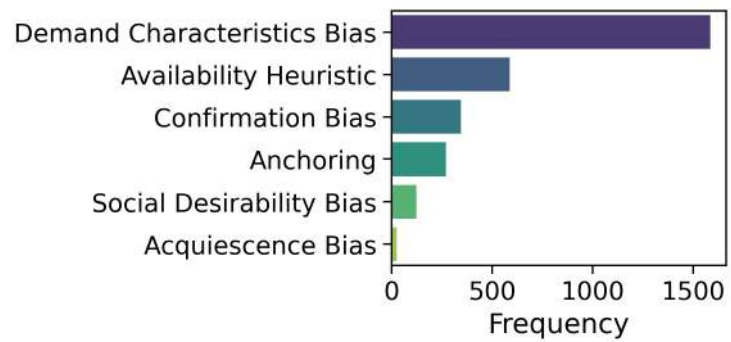
5.2.4 Cognitive Biases

We also report the frequency of the six cognitive biases that the large language model detects in the poll framing. Figure 5.5 shows the counts per cycle.

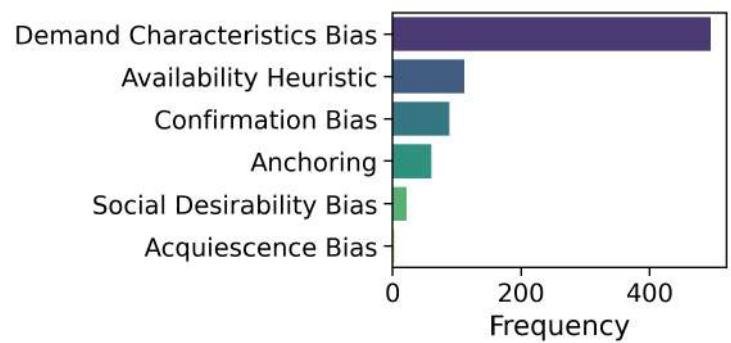
The cognitive-bias profile is stable across cycles. Demand Characteristics Bias dominates in every cycle, at roughly double the next most frequent bias, with the absolute counts tracking the dataset size (about 145 in 2016, 1,580 in 2020 and 490 in 2024). The Availability Heuristic, Confirmation Bias and Anchoring follow, and Social Desirability and Acquiescence Bias are rare. The dominance of Demand Characteristics Bias matches the expectation that respondents to a public poll answer towards what they read the author as wanting to hear.



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.5. Frequency of the six detected cognitive biases per cycle. Demand Characteristics Bias dominates in every cycle and the ranking of the remaining biases is stable across cycles.

5.3 Correlation Structure of the Bias Markers

We next report how the three structural markers relate to the author bias, the audience bias and the poll outcome. Figure 5.6 shows the Pearson correlation matrix per cycle, with significance after false-discovery-rate (FDR) correction. Table 5.5 collects the coefficients.

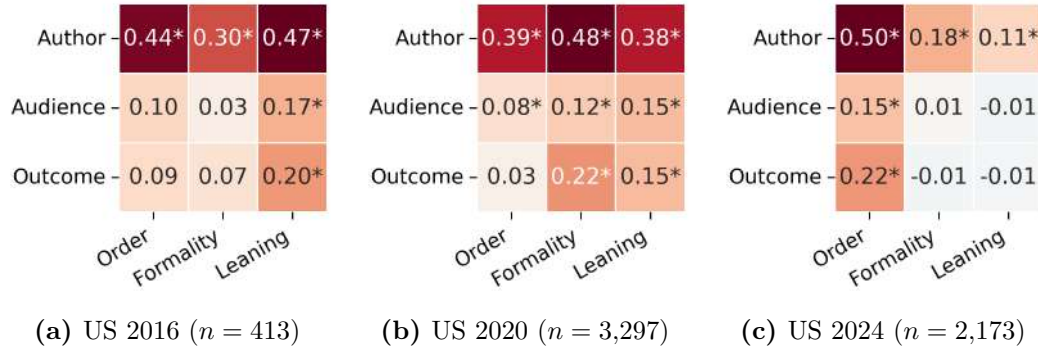


Figure 5.6. Pearson correlation between the three structural markers (columns) and the author bias, audience bias and poll outcome (rows). Cells marked with an asterisk are significant after FDR correction. The author-bias row is dark in every cycle; the outcome row weakens as the cycles progress.

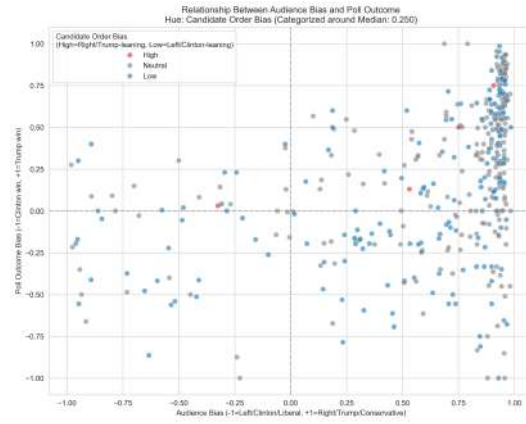
		Candidate order	Formality bias	Political leaning
US 2016	Author bias	0.440*	0.298*	0.466*
	Audience bias	0.100	0.035	0.166*
	Poll outcome	0.091	0.067	0.201*
US 2020	Author bias	0.390*	0.485*	0.382*
	Audience bias	0.083*	0.124*	0.151*
	Poll outcome	0.028	0.219*	0.154*
US 2024	Author bias	0.498*	0.175*	0.108*
	Audience bias	0.146*	0.009	-0.013
	Poll outcome	0.221*	-0.012	-0.011

Table 5.5. Pearson correlation coefficients between the structural markers and the three bias quantities. An asterisk marks significance after FDR correction.

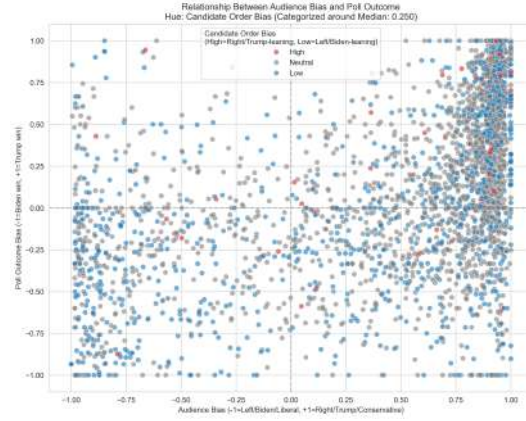
Three patterns emerge. First, the markers correlate most strongly with author bias in every cycle: the top row of each matrix is uniformly significant and carries the largest coefficients (up to 0.485 for formality bias in 2020 and 0.498 for candidate order in 2024). The markers therefore capture how biased the poll author is more

reliably than they capture audience behaviour or the outcome. Second, the link to the poll outcome changes character across cycles. In 2016 only political leaning predicts the outcome (0.201*); in 2020 formality bias (0.219*) and political leaning (0.154*) predict it while candidate order does not (0.028, not significant); in 2024 the pattern inverts and only candidate order predicts the outcome (0.221*) while formality and leaning collapse to near zero. Third, the count of significant cells peaks in 2020 (eight of nine) and is lower in 2016 and 2024 (five of nine each), which mirrors the accuracy ranking.

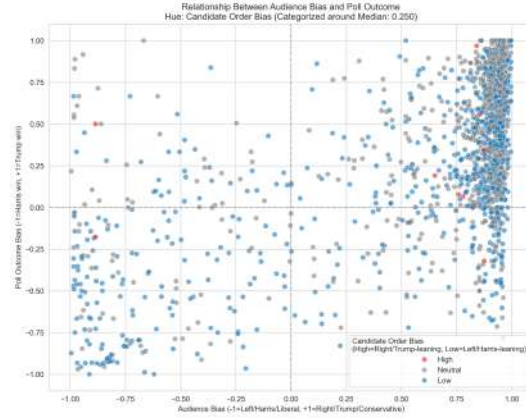
The scatter plots of Figures 5.7–5.9 display the same relationships at the level of individual polls. Each plots the audience bias (horizontal) against the poll outcome (vertical), with the points coloured by one marker. The dominant feature in every cycle is the dense cluster at the right edge: most poll audiences sit at the conservative pole and produce Trump-winning outcomes, which reproduces the audience skew of Section 5.1. Within that cluster the marker colour separates the points only weakly, consistent with the modest outcome correlations in Table 5.5.



(a) US 2016

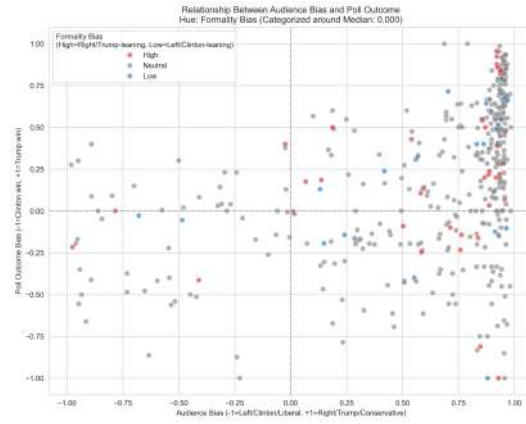


(b) US 2020

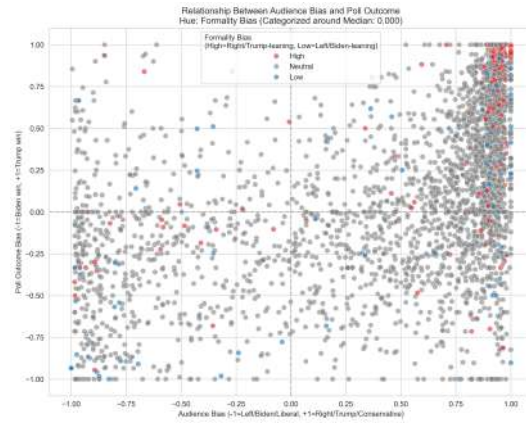


(c) US 2024

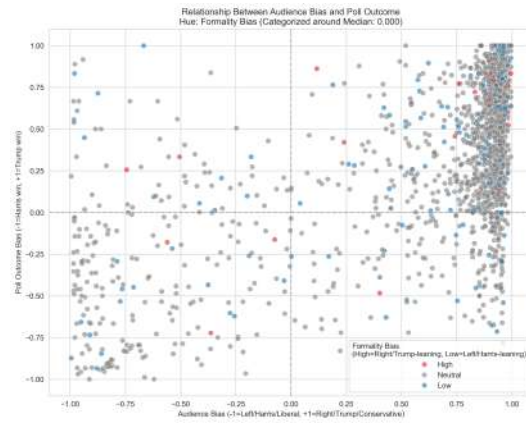
Figure 5.7. Audience bias against poll outcome, coloured by candidate order bias. The point cloud is concentrated at the conservative pole in every cycle.



(a) US 2016

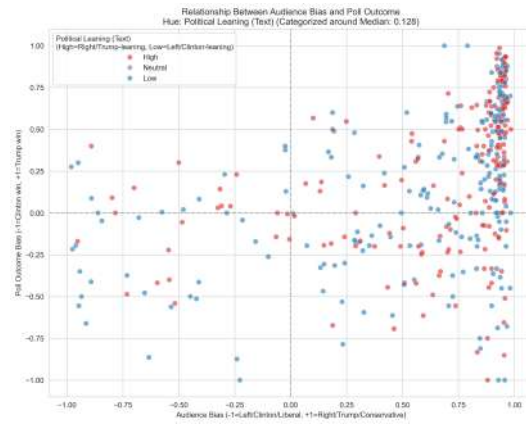


(b) US 2020

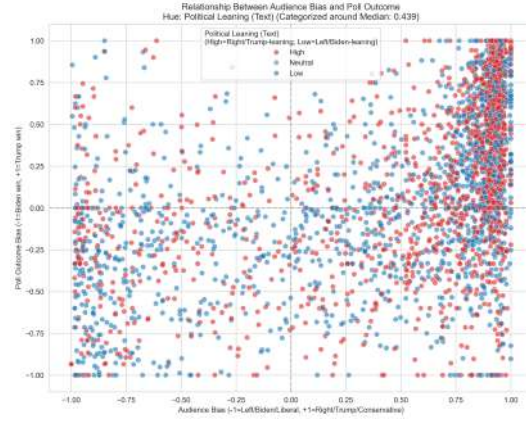


(c) US 2024

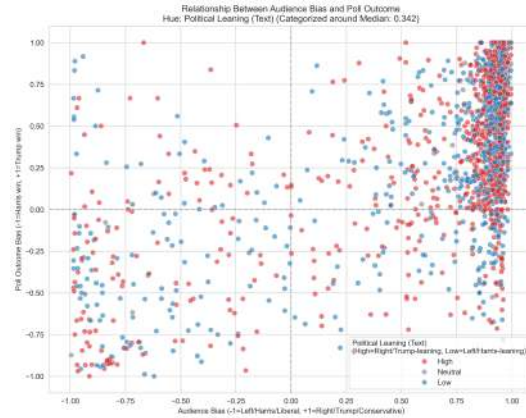
Figure 5.8. Audience bias against poll outcome, coloured by formality bias. The formality colour shows no clear spatial separation in any cycle.



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.9. Audience bias against poll outcome, coloured by the textual political leaning. Pro-Trump and pro-opponent texts mix throughout the cloud, most thoroughly in 2024.

5.4 Model Specification and Fit

We now ask whether adding the bias markers improves the model. We compare the full SCWG against a baseline that drops the structural markers, by two criteria: the binomial GLM that produces the estimates and an ordinary least squares (OLS) model fitted for diagnostics. Lower AIC (Akaike Information Criterion) and higher log-likelihood favour a model under the GLM; higher adjusted R^2 and lower AIC favour it under OLS. Tables 5.6 and 5.7 report the comparison.

Cycle	Metric	Full	Baseline	Improvement
US 2016	AIC	158,946	162,249	3,303
	Log-likelihood	−79,450	−81,118	+1,668
US 2020	AIC	3,339,813	3,509,916	170,103
	Log-likelihood	−1,669,883	−1,754,952	+85,070
US 2024	AIC	5,529,792	5,583,809	54,017
	Log-likelihood	−2,764,872	−2,791,899	+27,027

Table 5.6. GLM comparison of the full SCWG against the baseline without bias markers. The full model wins on both criteria in every cycle.

Cycle	Full adj. R^2	Baseline adj. R^2	Improvement
US 2016	0.1195	0.0903	+0.0292
US 2020	0.2140	0.1715	+0.0426
US 2024	0.3720	0.3428	+0.0292

Table 5.7. OLS adjusted R^2 for the full and baseline models. The bias markers add explanatory power in every cycle.

The bias markers improve the model in every cycle, on every criterion. The full GLM has the lower AIC and the higher log-likelihood throughout, and the full OLS has the higher adjusted R^2 . The size of the gain varies. The AIC improvement is far larger in 2020 (about 170,000) than in 2016 (about 3,300), which reflects both the larger sample and a stronger, more consistent bias signal in 2020. The adjusted R^2 rises from cycle to cycle (0.12, 0.21, 0.37), so the markers explain the most variance in 2024. We read this carefully in Section 5.5: a high R^2 in 2024 means the markers describe the recorded outcome well, not that the correction reaches the truth.

We check multicollinearity through the variance inflation factor (VIF) of the OLS model, flagging any value above 5.0. No marker breaches the threshold in any cycle.

The factors stay below 2.0 in 2016 and 2020, with the highest values among the sentiment and oppose-score features (for example, `sentiment_intensity` at 1.83 in 2016 and `candidate_A_oppose_score` at 1.94 in 2020). In 2024 the factors rise but remain acceptable, peaking at 2.93 for `candidate_A_oppose_score` and near 2.7 for the sentiment features, which signals mild collinearity among the sentiment-related predictors. One indicator, `bias_acquiescence`, returns an undefined VIF in 2016 because it varies too little in that cycle. The estimates are therefore stable, though we read the individual sentiment coefficients in 2024 with care.

5.5 Corrected Estimates and Accuracy

We now read the accuracy of the post-stratified estimate. The full SCWG corrects the raw signal towards the truth in every cycle, but by a margin that shrinks as the sample skew grows.

	US 2016	US 2020	US 2024
	(Trump/Clinton)	(Trump/Biden)	(Trump/Harris)
Analysis-ready polls	759	7,386	2,724
Vote-weighted mean (%)	65.47	64.05	80.28
Full SCWG estimate (%)	59.18	48.85	73.08
Official result (%)	48.90	47.70	50.60
SCWG deviation (pp)	+10.28	+1.15	+22.48

Table 5.8. Headline vote-share estimates per cycle, expressed as the positive-direction (Trump) two-way share. The deviation is the full SCWG estimate minus the official result, in percentage points (pp). The SCWG estimate is closest to the truth in 2020 and furthest in 2024.

In 2020 the correction is near-exact. The vote-weighted mean overestimates Trump at 64.05%; the full SCWG brings this down to 48.85%, within 1.15 pp of the 47.70% result, a reduction of more than 15 pp. This is the best-performing cycle. In 2016 the correction is partial. The vote-weighted mean of 65.47% falls to 59.18%, which still sits 10.28 pp above the 48.90% result: the model removes roughly a third of the overestimation. In 2024 the correction is weakest. The vote-weighted mean of 80.28%, the most extreme of the three cycles, falls only to 73.08%, which remains 22.48 pp above the 50.60% result. The model removes about 7 pp of an overestimation that exceeds 29 pp.

The accuracy ranking tracks the audience skew of Section 5.1 and the correlation

count of Section 5.3. Post-stratification reweights the strata the model has seen; it cannot recover a stratum that the sample barely contains. By 2024 the audience sits almost entirely at the conservative pole, so the liberal strata carry few observations and the reweighting has little to act on. The model in 2024 fits the recorded outcome well, as the high adjusted R^2 of Table 5.7 shows, yet it cannot close the gap to the electorate, because the gap is one of representation rather than of framing.

5.6 Temporal Accuracy and External Benchmarking

The estimates above are single, campaign-wide figures. We also apply the fitted SCWG to rolling 7-day windows to obtain a daily time series of the bias-adjusted share, as Chapter 4 describes. The project ships an interactive dashboard for each cycle in the repository¹, which overlays the raw, baseline and full SCWG series, the prediction-market price and vertical markers for campaign milestones such as debates and conventions, and lets the reader hover and zoom. The thesis reproduces this dashboard as static exports in three pairings: the SCWG against the raw vote-weighted baseline, the SCWG against the baseline SCWG and the SCWG against the prediction market. We benchmark the daily series against the official result through the mean absolute error (MAE) and against the PredictIt prediction market, which we read as an implied win probability renormalised to a two-way share [5, 26]. Table 5.9 reports the errors.

Method (MAE, pp)	US 2016	US 2020	US 2024
Raw vote-weighted	18.73	27.71	28.09
Baseline SCWG	15.13	10.10	23.47
Full SCWG	11.44	4.28	21.82
PredictIt market	17.08	10.60	4.06
Daily data points	147	292	107

Table 5.9. Mean absolute error of each daily series against the official result, in percentage points. The full SCWG is the most accurate Twitter-based series in every cycle. The prediction market is the most accurate series overall in 2024.

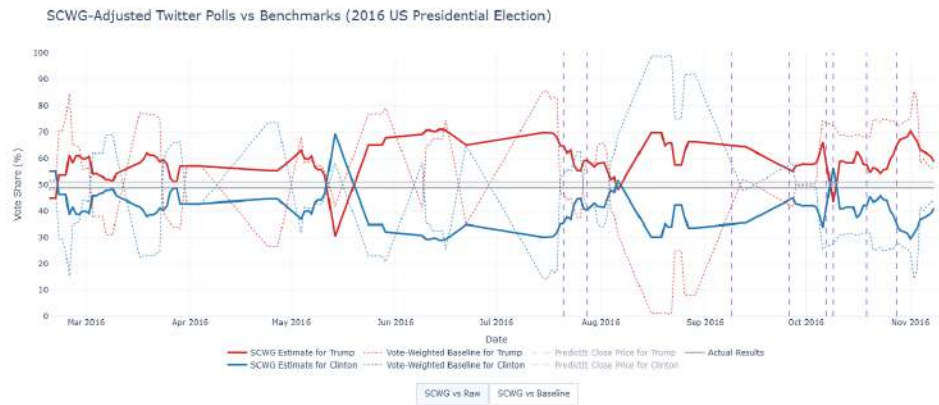
The full SCWG is the most accurate Twitter-based series in every cycle. It improves on the raw vote-weighted series by 7.30 pp in 2016, 23.44 pp in 2020 and 6.27 pp in 2024, and on the baseline SCWG by 3.69 pp, 5.82 pp and 1.65 pp respectively. The improvement over the baseline is therefore largest in 2020 and smallest in 2024,

¹<https://github.com/mivige/TweetPollBias>

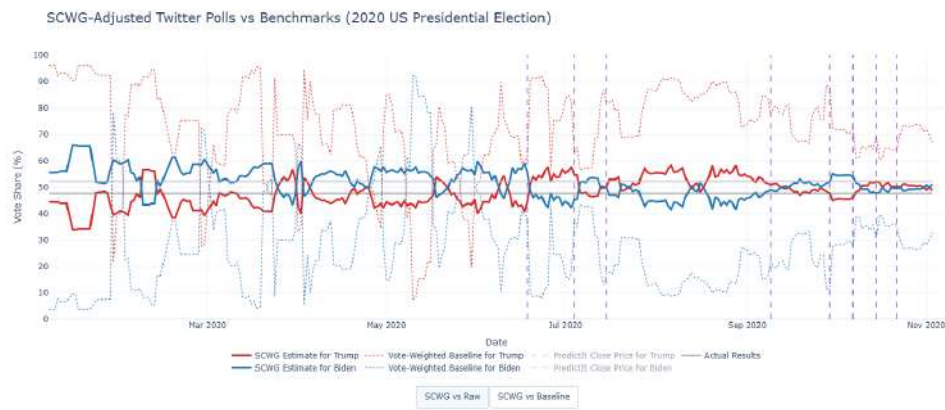
which restates the model-fit pattern of Section 5.4: the markers add the most when the sample still carries a correctable signal and the least when the audience skew saturates the outcome.

Figures 5.10–5.12 show the three static pairings. Against the raw vote-weighted baseline (Figure 5.10), the SCWG series is visibly smoother and closer to the result line: the raw baseline swings between near 0% and above 85% throughout the campaign and is essentially uninterpretable, most dramatically in 2020, while the SCWG series tracks the result line closely. Against the baseline SCWG (Figure 5.11), the gap between the full and baseline series is small but consistent in 2016, clearly visible in 2020 where the full series often sits on the result line through the second half of the campaign and almost nil in 2024 where the two series run on top of each other far above the result line.

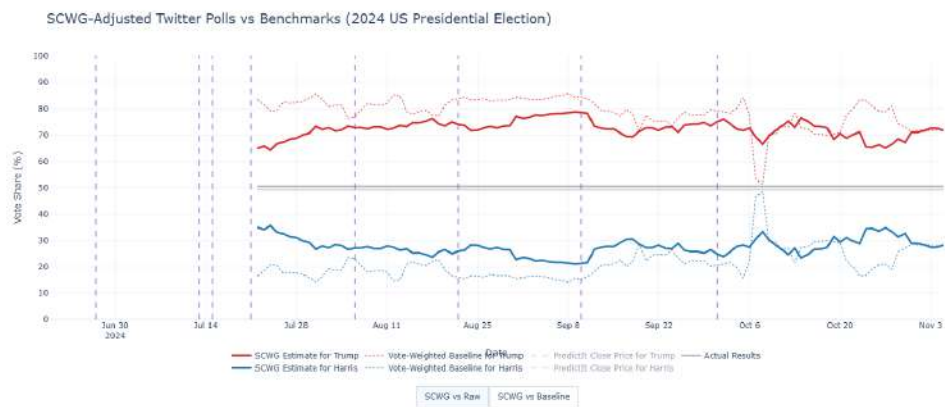
The comparison with the prediction market (Figure 5.12) inverts across cycles. In 2016 the market (MAE 17.08 pp) is less accurate than the SCWG (11.44 pp), and the mean divergence between the two is large, at 28.78 pp. In 2020 the two are closer, with the market at 10.60 pp and the SCWG at 4.28 pp, and a divergence of 12.94 pp; the market’s error comes mainly from its volatility during the early-2020 primary period, after which it stabilises near the result. In 2024 the market is the most accurate series of all, at 4.06 pp, while the SCWG sits at 21.82 pp and the two diverge by 23.67 pp. The market and the Twitter estimate agree most when the Twitter signal is usable and disagree most when it is not.



(a) US 2016

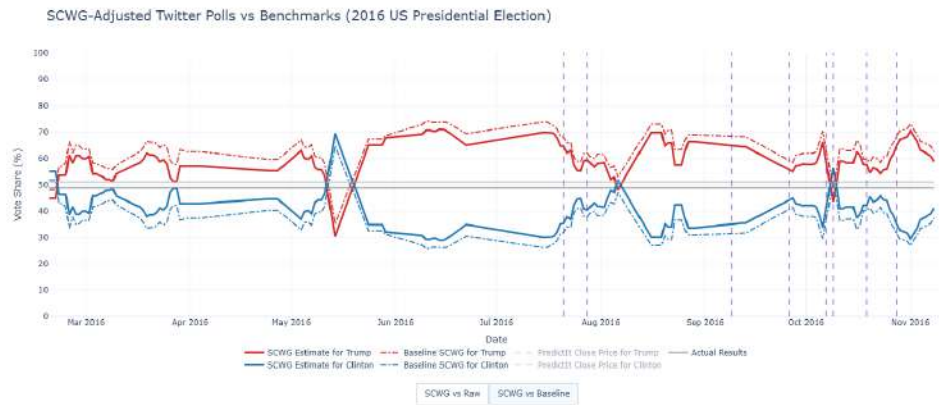


(b) US 2020

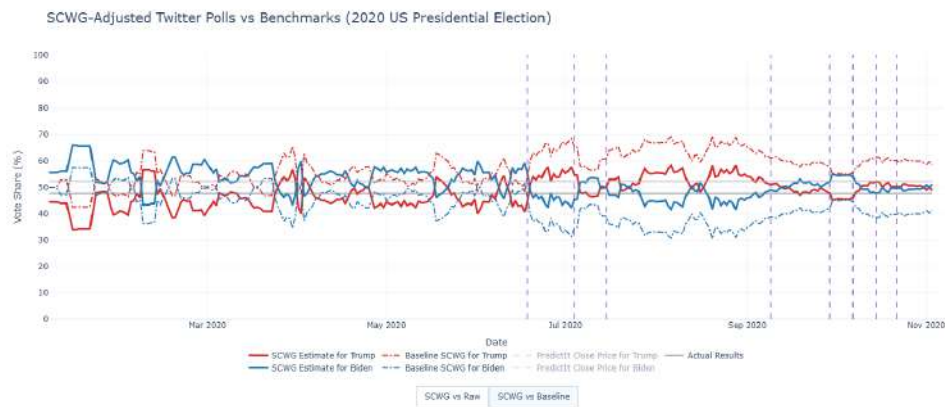


(c) US 2024

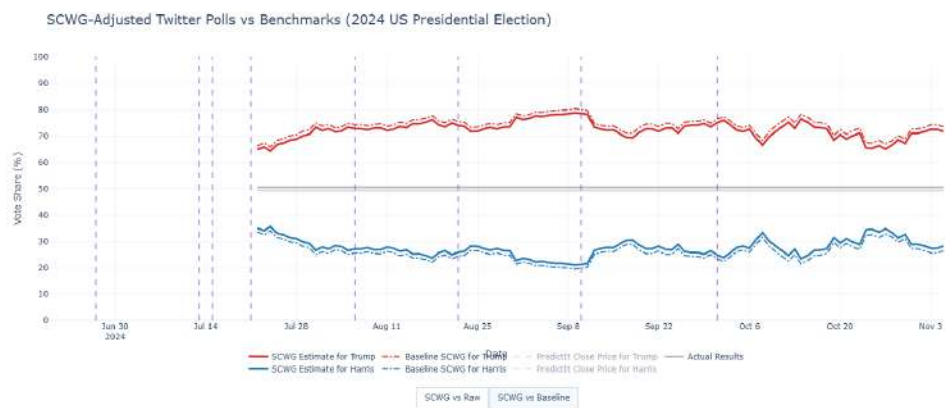
Figure 5.10. Static export of the dashboard: the full SCWG series against the raw vote-weighted baseline. Horizontal grey lines mark the official results and vertical dashed lines mark campaign milestones. The raw baseline is volatile in every cycle; the SCWG series is smooth and, in 2016 and 2020, close to the result line.



(a) US 2016

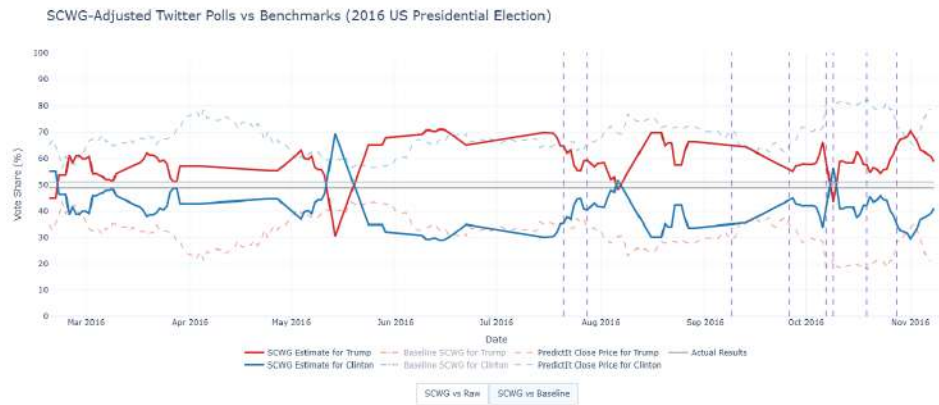


(b) US 2020

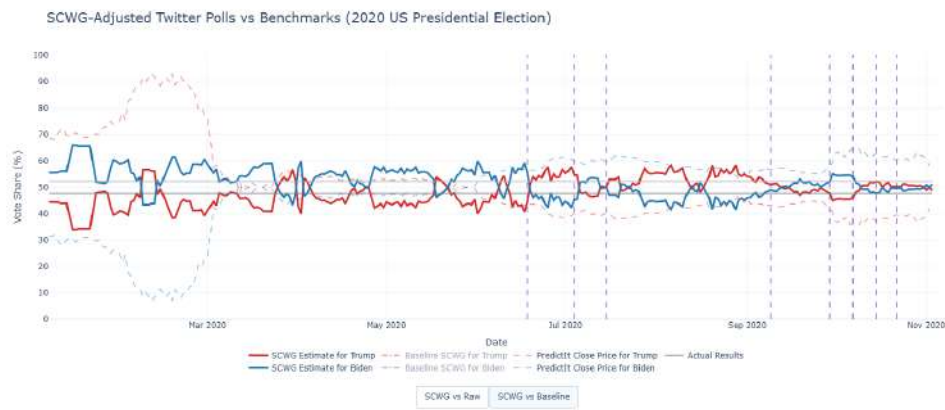


(c) US 2024

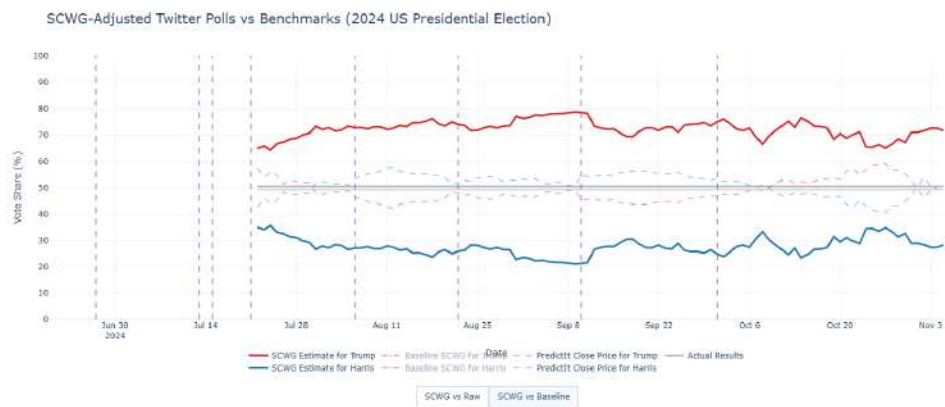
Figure 5.11. Static export of the dashboard: the full SCWG series against the baseline SCWG without bias markers. The bias markers shift the estimate downward; the shift is consistent in 2016, largest in 2020 and negligible in 2024.



(a) US 2016



(b) US 2020



(c) US 2024

Figure 5.12. Static export of the dashboard: the full SCWG series against the PredictIt prediction-market price. The market is comparable to or worse than SCWG in 2016 and 2020, but tracks the result closely in 2024 while the SCWG series sits about 20–25 pp too high.

5.7 Summary

The results form one coherent picture. The poll audience is unrepresentative in every cycle and grows more so over time (Section 5.1). The structural markers move the recorded vote share through primacy, naming and textual leaning, with effects that are real but candidate- and cycle-specific (Section 5.2), and they correlate most strongly with author bias rather than with the outcome (Section 5.3). Adding the markers improves the model on every criterion in every cycle (Section 5.4). The post-stratified estimate corrects the raw signal towards the truth, to within 1.15 pp in 2020, but leaves 10.28 pp in 2016 and 22.48 pp in 2024 (Section 5.5). Over the campaign the SCWG is the most accurate Twitter-based series throughout, yet the prediction market overtakes it in 2024 (Section 5.6). The recurring theme is that the correction succeeds in proportion to how much usable signal the sample retains. Chapter 6 interprets this theme.

Chapter 6

Discussion

This chapter interprets the results of Chapter 5. We begin with the central finding and what it means for the problem the thesis sets out to solve (Section 6.1). We then ask what the structural bias markers contribute to the correction (Section 6.2) and we read the whole picture through the representation-versus-measurement split that organises the thesis (Section 6.3). We discuss the prediction-market benchmark and what it implies about the platform signal (Section 6.4). We then place the work against the prior literature and state its contribution plainly (Section 6.5), and we close with the caveats that bound the reading of the model (Section 6.6).

6.1 Principal Findings

The framework corrects the social-media poll signal towards the electorate and the size of the correction it achieves depends on how representative the sample still is. After the head-to-head model and the counterfactual post-stratification, the full SCWG estimate sits 1.15 pp above the official result in 2020, 10.28 pp above it in 2016 and 22.48 pp above it in 2024 (Table 5.8). The 2020 figure is an accurate recovery from a raw signal that overstated Trump by more than 16 pp. The 2016 figure is a partial correction, removing about a third of the overestimation. The 2024 figure is a weak correction of an extreme starting point, removing about 7 pp of a gap above 29 pp.

We do not present the 2024 number as a success, and the honest reading is the informative one. The three cycles are not three independent tests of the same problem; they are three points on a worsening trend. The audience distribution of Section 5.1 moves further towards the conservative pole at each cycle, until in 2024

almost the entire sample sits at the extreme. The accuracy of the correction tracks that trend exactly. This tells us what the method can and cannot do: it can reweight the strata the sample contains, it cannot manufacture a stratum the sample has all but lost. A change in the platform itself accompanies this trend, and Section 6.3 argues that it, rather than the model, sets the limit in 2024.

6.2 What the Bias Markers Contribute

The bias markers earn their place in the model. They lower the AIC and raise the log-likelihood of the GLM in every cycle, and they raise the adjusted R^2 of the OLS model in every cycle (Tables 5.6 and 5.7). The markers add real, repeatable explanatory power; they are not noise.

Their contribution to the corrected estimate is more measured and it follows the same trend as the headline accuracy. Over the campaign, the markers improve on the baseline SCWG by 3.69 pp in 2016, 5.82 pp in 2020 and only 1.65 pp in 2024 (Table 5.9). The markers help most in 2020 and least in 2024. The correlation structure of Section 5.3 explains why. In every cycle the markers correlate most strongly with author bias, not with the poll outcome: they describe who writes a biased poll more reliably than they shift the result. Their link to the outcome is the part that matters for the estimate and that link erodes across cycles. By 2024 the textual leaning carries no outcome signal at all, because Trump's recorded share is flat across pro-Trump, neutral and pro-opponent text (Table 5.4); only candidate order still predicts the outcome. When the audience is this homogeneous, the framing of a poll can no longer move its result, so a marker that measures framing has little left to correct.

There is one reading of 2024 to set aside. The OLS adjusted R^2 is highest in that cycle, at 0.37 (Table 5.7). This does not mean the correction works best in 2024. A high R^2 means the model explains the variance in the recorded outcome, and in 2024 the recorded outcome is well explained precisely because the structural skew is strong and consistent. The model describes the biased sample faithfully; describing it faithfully is not the same as recovering the electorate behind it.

6.3 Representation versus Measurement Error

The total-error perspective of Chapter 2 [20] separates two kinds of error: representation error, which concerns who is in the sample, and measurement error, which concerns what the instrument records. The thesis corrects each with a different tool.

The population weights correct representation, by reweighting the strata to their true share of the electorate. The zero-bias counterfactual corrects measurement, by stripping the framing biases out of the prediction while it keeps each stratum's genuine composition. The results show that these two corrections are not interchangeable and that one cannot stand in for the other.

The 2020 cycle is the case where both corrections have something to work with. The sample is skewed but still contains enough of each stratum for the weights to act on and the framing markers still carry an outcome signal for the counterfactual to remove. Both corrections fire and the estimate lands within 1.15 pp. The 2024 cycle is the case where the representation error is so large that the measurement correction cannot compensate for it. When the audience sits almost entirely at one pole, the liberal strata are nearly empty, and reweighting an empty stratum recovers nothing. Removing the framing bias from a sample that has no one left to reweight towards leaves the estimate where it was. The measurement-side tool is sound, but it operates on the wrong axis to fix a representation failure of this size.

A structural change in the data source accompanies this collapse. Between the 2020 and 2024 cycles, Twitter changed ownership and was rebranded as X, and its feed algorithm was retuned in ways that amplify divisive and partisan content, an effect that Gauthier et al. [6] document directly. This matters for post-stratification beyond the audience skew of Section 5.1, because it threatens the assumption the method rests on. Post-stratification assumes that, within a stratum, a user on the platform behaves like the corresponding member of the electorate: that a liberal Democrat in the sample answers a poll as a liberal Democrat in the population would. The 2024 raw signal breaks that assumption. A vote-weighted baseline near 80% for Trump implies that even polls written by left-leaning authors are answered overwhelmingly in his favour, which is consistent with partisan amplification and coordinated participation rather than with the genuine preference of those strata. When the within-stratum behaviour is itself distorted, reweighting the strata to their true population shares cannot undo the distortion: the cells are correct in size but wrong in content. This is why 2024 resists correction in a way that 2016 and 2020 do not, and it locates the failure in the platform rather than in the model.

This is the substantive lesson of the thesis. A correction method for social-media polls is only as good as the representativeness the sample retains. The two-sided correction adds value while both error sources are moderate and the representation side sets the ceiling once the sample collapses towards one pole.

6.4 The Prediction Market and the Platform Signal

The prediction-market benchmark of Section 5.6 sharpens the same point from the outside. The PredictIt series is less accurate than the SCWG in 2016 and 2020, and far more accurate in 2024 (Table 5.9). The market draws on information beyond the platform, while the SCWG draws only on the polls. The two agree when the platform signal is usable and diverge when it is not. The 2024 divergence of 23.67 pp is the clearest case: the market tracks the roughly even race while the Twitter estimate stays 20–25 pp too high throughout the campaign.

We read this as a statement about the data source rather than about the method. A correction built on the platform’s own polls inherits the platform’s blind spots. When the platform’s user base diverges far enough from the electorate, as in 2024, no amount of on-platform reweighting recovers the missing voters and an off-platform aggregator carries the stronger signal. The prediction market is not a competitor to the SCWG so much as a reminder of the source’s limits.

6.5 Relation to Prior Work and Contribution

We state our relation to the prior work plainly, to avoid overclaiming. Post-stratifying a non-representative online sample is established practice. Wang et al. [24] recover accurate forecasts from a heavily skewed online sample and the Social Polls line that supplies our data already studies and corrects social-media election polls [21, 17, 18]. The idea of fitting a model on a skewed sample and reweighting its predictions to the population is not ours and the demographic inference that supports it is taken as given from external pipelines [1, 25]. A reader familiar with this line will reasonably ask whether the post-stratification of social polls has been done before. It has.

The contribution of this thesis lies in two additions to that established frame, not in the frame itself. First, we integrate the measurement-side framing biases of Chapter 2 directly into the correction as bounded markers: the candidate order M_1 , the appellative formality M_2 and the textual leaning M_3 , alongside sentiment, toxicity and the cognitive biases. Prior social-poll work treats these framing choices as biases to document; we encode them as predictors inside the model that produces the estimate. Second, we add the zero-bias counterfactual, which projects the estimate under a neutral polling environment while it preserves each stratum’s authentic partisan and ideological identity. This separates the framing distortion that the poll creator introduces from the genuine composition of the electorate and corrects only the former.

We also depart from the canonical method on the modelling side. The standard instance of this idea is multilevel regression and post-stratification (MRP), which pairs a hierarchical model with a post-stratification grid [7, 15]. We do not use MRP. We fit a single-level sample-calibrated weighted GLM, weighted by each poll’s vote count, and post-stratify its predictions. The contribution is therefore the integration of measurement-side markers and the zero-bias counterfactual within a single-level weighted GLM, applied uniformly across three cycles through the election-agnostic pipeline of Chapter 4.

6.6 Reading the Coefficients: Collinearity and Scope

Two caveats bound how far the individual coefficients can be read. First, multicollinearity. The variance inflation factors of Section 5.4 stay within the conventional threshold of 5.0 in every cycle, so the estimates are stable and we retain all markers. The factors rise in 2024, however, peaking near 2.9 among the sentiment and oppose-score features, which signals mild collinearity in that group. We therefore read the aggregate estimate with confidence but treat the individual sentiment coefficients in 2024 with care and we do not interpret any single coefficient as an isolated causal effect.

Second, scope. The marginal effects of Section 5.2 are descriptive: they mix the framing effect of a marker with the genuine preference of the audience that the marker happens to co-occur with. The model separates the two for the estimate, but the per-level tables should be read as associations, not as the causal effect of renaming or reordering a poll. The framing markers also depend on inference pipelines and external tools taken as given and the population weights rest on coarse exit-poll strata. We collect these design choices and their consequences in Chapter 7.

6.7 Summary

The discussion reduces to one argument. The two-sided correction works in proportion to the signal the sample retains. It recovers the 2020 electorate almost exactly, corrects 2016 partially and barely moves the 2024 estimate, because the 2024 sample has collapsed towards one pole and no reweighting can restore the missing side. The bias markers improve the model in every cycle and help the estimate most when the framing still moves the outcome, which is to say least in 2024. The prediction market confirms from the outside that the limit is the data source rather than the method. The contribution is the integration of measurement-side framing markers and a zero-bias counterfactual into the established practice of post-stratifying social

polls, through a single-level weighted GLM rather than MRP. Chapter 7 turns these readings into the explicit limitations of the design and the lines of work they open.

Chapter 7

Limitations and Future Work

This chapter sets the boundaries of the work. We have two aims. First, we state the limitations of the design and we justify each decision as reasonable given the data and the goal, so the reader can judge how far the estimates travel. Second, we turn each limitation into a line of work that the findings open, with direct reference to the result that motivates it.

We organise the limitations by where they enter. Section 7.1 covers the data and its source. Section 7.2 covers the method. Many of the method limitations share a direction: where each one acts, it causes the correction to under-shoot, so it leaves the positive-direction (Trump) estimate above the truth rather than below it. This matters because it is the same direction as the residual bias of Chapter 5: these limitations help explain why that residual is one-sided and why it grows with the sample skew, rather than averaging out across cycles. Section 7.3 then describes the future work.

7.1 Data and Source Limitations

The data carry three limitations that originate in how they are collected rather than in how we model them. We take each as an inherited property of the source.

Keyword-based sampling. The poll corpus comes from a keyword query: the tokens `vote` or `voting` together with the candidate names of each cycle (Chapter 3). A poll that asks about the election without these tokens does not enter the sample, so the corpus under-represents polls that frame the contest in other words. We accept this because the keyword query is what makes a large collection tractable and

comparable across cycles, and because the same query underlies the prior corpora that we extend [17, 18]. The cost is that our sample is a keyword-defined slice of the poll population rather than the whole of it.

Public versus private vote counts. We observe the public vote tally of each poll. Scarano et al. [18] document a systematic gap between this public tally and a private count available only to the poll author, which excludes purchased or otherwise inauthentic votes. We cannot observe the private count, so any inauthentic votes in the public tally pass into our outcome variable Y_i unmeasured. This gap belongs to the platform’s reporting, not to our pipeline, and we inherit it. It bears on the 2024 cycle in particular, where the audience sits almost entirely at one pole (Section 5.1).

Overlap in the US 2020 sources. The US 2020 data draw on three overlapping sources, of which the decahose sample is the largest and captures many polls immediately after publication, before they accumulate votes (Table 3.2). This overlap inflates the raw poll count for 2020 relative to the other two cycles. We remove the duplicates and the empty polls at load time, so the analysis set is clean, but the raw counts of that cycle are not directly comparable with 2016 and 2024.

7.2 Method Limitations

The method carries a longer list. We group it by the stage at which each limitation enters and we begin with the one that sets the ceiling on the whole correction.

Failure of within-stratum exchangeability. Post-stratification rests on an exchangeability assumption: within a stratum, a user on the platform answers a poll as the corresponding member of the electorate would (Section 6.3). The correction reweights the strata to their true population share, but it takes the behaviour inside each stratum as representative. The 2024 cycle breaks this assumption. A vote-weighted baseline near 80% for the positive-direction candidate (Table 5.8) implies that even the strata we would expect to lean the other way answer in his favour, which is consistent with the partisan amplification and coordinated participation that the retuned platform feed promotes [6]. We infer this distortion from the raw baseline; we do not measure it. The pipeline performs no bot or coordination detection, so we describe the pattern as consistent with amplification rather than as established brigading. Where the assumption fails, the cells of the grid are correct in size but wrong in content, and no reweighting recovers a preference that the sample no longer contains. This limitation bounds every estimate and it is most severe in 2024 (Section 5.5).

Circular quantile anchoring. We set the continuous stratum predictors to in-sample quantiles of the Twitter distribution: the 75th percentile for the Republican stratum, the 25th for the Democrat stratum and the median for the Independent stratum (Chapter 3). This anchors the post-stratification frame to the same skewed sample that the correction tries to fix. We make the choice deliberately. Absolute population values lie outside the range the model has seen, and predicting there forces the fitted surface to extrapolate, which destabilises the estimate (Section 4.2.3 gives the full rationale). Anchoring to in-sample quantiles keeps every stratum inside the training support, so the model interpolates. The cost is that the correction is scoped to within-platform skew: it repositions the strata relative to one another on the platform’s scale, but it does not lift the sample onto the electorate’s scale.

Vote-count weighting. The model weights each poll by its two-candidate vote total, $W_i = N_i$ (Section 4.2.1). This mirrors standard survey practice, where a larger sample carries a tighter variance and earns more weight. It has a side effect. A poll that goes viral or attracts coordinated participation carries more weight precisely because it is large, and these are the polls most exposed to the amplification described above. The weighting is sound for binomial variance, but in a cycle where the largest polls are also the most distorted it is a channel through which the distortion enters the estimate.

Neutral-zero imputation. When the merge from polls to inferred variables leaves a bias marker missing, we impute the neutral value zero (Section 4.2.1). The choice is conservative: it assumes no bias where we have no evidence of bias. It also pushes the marker distribution towards neutrality whenever the upstream inference pipeline has gaps, so it understates the framing signal for the affected polls and the correction does less work, which is again in the direction of leaving the estimate too high.

Coarse strata and population-level ideology. The post-stratification grid crosses a three-way partisan partition with a three-way ideological partition, giving nine strata (Section 3.3). The partisan dimension is inferred per user; the ideological dimension is not, because no per-user ideology pipeline is available in our data. Ideology therefore enters only at the population level, through exit-poll shares, and acts on the model through a fixed multiplier per ideological group rather than a value inferred for each user. Nine cells cannot capture the joint distribution of the electorate in fine detail, and the fixed multiplier is an approximation rather than a measurement.

Relative position rather than an absolute scale. Following the same logic as the anchoring choice, we keep the relative position of each stratum on the platform’s partisan scale and we do not map that scale onto an absolute real-world scale. The feed-driven distortion makes such mapping unreliable, since we have no fixed point on the platform that we can trust to correspond to a fixed point in the electorate. The consequence is that the estimate is a corrected platform quantity, not a direct measurement of population preference, and the reader should treat it as such.

External inference pipelines taken as given. The author partisanship, the audience partisanship and the author demographics come from external pipelines: Bayesian ideal-point estimation [1] for partisanship and the M3-Inference model [25] for gender and age. We take their outputs as given and we neither re-estimate nor validate them on our data, so any error in these pipelines propagates into our controls. We accept this because re-estimating either pipeline is a project in itself and because both are established tools, but the validity of our controls is conditional on theirs. Unlike the limitations above, this one adds uncertainty without a direction we can sign.

Off-the-shelf markers and an unvalidated language model. Several markers depend on off-the-shelf tools with their own error rates. Named-entity recognition uses the spaCy `en_core_web_sm` pipeline [14] and entity disambiguation uses DBpedia Spotlight (Section 4.1); either can mislabel a span. The cognitive-bias markers come from a large language model (`gemini-3.1-flash-lite`) prompted to act as a cognitive scientist (full prompt in Appendix B.1), and we do not validate its labels against human annotation. We report the cognitive-bias frequencies as descriptive (Section 5.2.4) for this reason.

An off-platform stance lexicon. The interpersonal-stance lexicon that we build on [16] was developed on Reddit rather than on Twitter. Language conventions differ between the two platforms, so the lexicon may transfer imperfectly to poll text. We use it as one signal among several rather than as a sole measure, which limits the exposure to this mismatch.

These limitations differ in stage and in severity. The first four share a direction: each causes the method to under-correct and so leaves the positive-direction estimate above the truth. The remainder add uncertainty around the estimate without a sign we can fix. Together they explain why the residual bias of Chapter 5 is one-sided and why it tracks the sample skew, near-exact in 2020 and largest in 2024.

7.3 Future Work

Each limitation points to a line of work. We order them by how directly they address the central failure.

External ground-truth calibration. The strongest line follows from the anchoring limitation. The correction is scoped to within-platform skew because the stratum predictors are anchored to the platform’s own distribution. A method that also anchored the *content* of each stratum, not only its size, to an external signal would break this circularity. One could calibrate the stratum predictors against concurrent off-platform measurements from the same period, such as established survey outcomes or other independent estimates, so that the frame no longer depends solely on the sample it corrects. This is the principled fix for the 2024 failure (Section 5.5): it would let the method reach a stratum that the platform sample has all but lost, which reweighting alone cannot do.

Bot and coordination detection. The headline limitation rests on an inference: we read the 2024 raw baseline as consistent with amplification and coordinated participation, but we do not measure them. Adding bot and coordination detection upstream of the pipeline would test this hypothesis directly. It would also let the model flag or down-weight distorted polls explicitly, rather than absorbing them through the vote-count weighting.

Extending the configuration registry. The pipeline is election-agnostic by design. Each cycle is described by a registry entry that supplies the candidate names, their variants and the population weights, while the extraction and modelling logic stays fixed (Chapter 4). This makes it straightforward to extend the framework to new presidential cycles, to down-ballot races, or to other platforms that host polls. Each extension is a new registry entry rather than new code, and it would test how far the findings generalise beyond the three cycles studied here.

Porting the markers into an established MRP pipeline. Our model-fit comparison measures what the markers add inside the SCWG (Tables 5.6 and 5.7), not what they add to the established practice. Scarano et al. [17] post-stratify the same Twitter polls through a multilevel regression and post-stratification (MRP) model and report errors as low as one to two percentage points (Chapter 2). A clean test of our contribution is to insert the measurement-side framing markers and the zero-bias counterfactual into that MRP pipeline, holding the model family fixed, and to measure the change in accuracy. This separates the contribution of the

markers from our choice of a single-level weighted GLM over MRP (Section 6.5), and it reports how much the markers improve a strong existing baseline rather than our own.

Validating the markers. Finally, the cognitive-bias and textual-leaning markers would benefit from validation against human annotation. A sample of polls coded by hand for the six cognitive biases and for textual leaning would let us measure the precision of the language model and of the zero-shot classifier, and would put the marker-level readings of Chapter 5, such as the collapse of the leaning signal in 2024 (Table 5.4, Section 5.2.3), on firmer ground. The same labelled sample would support the mean-absolute-error comparison of Table 5.9 with a validated rather than an inferred set of markers. A sensitivity analysis over the ideology multipliers would also strengthen the design: sweeping the Conservative and Liberal multipliers over a range around the chosen values and reporting how the post-stratified estimate moves would show whether the headline results are robust to this assumption, and an external calibration signal could replace the a-priori values entirely.

Chapter 8

Conclusions

This thesis set out to measure and correct the structural biases that distort election polls on social media. We built an election-agnostic pipeline that encodes the framing of a poll as bounded markers, folds them into a Sample-Calibrated Weighted Generalized Linear Model (SCWG) and post-stratifies its predictions under a zero-bias counterfactual. We applied the same pipeline to the US 2016, 2020 and 2024 presidential elections. One finding runs through the results: the correction works in proportion to the representativeness the sample retains.

The three cycles are not three independent tests of the method; they are three points on a worsening trend. The poll audience drifts further towards the conservative pole at each cycle, and the accuracy of the correction tracks that drift. The recovery is near-exact in 2020, partial in 2016 and weak in 2024. Post-stratification can reweight the strata that the sample contains, but it cannot rebuild a stratum that the sample has lost. By the most skewed cycle the audience sits almost entirely at one pole, so the reweighting has little left to act on.

The framing markers earn their place. They improve the model in every cycle and they capture how biased a poll’s author is more reliably than they capture its outcome. Their leverage over the result, which is the part that matters for the estimate, erodes as the sample skew grows: by 2024 the textual leaning of a poll no longer moves its outcome, because the audience composition overwhelms it. The two-sided correction, population weights for representation and the counterfactual for measurement, adds value while both error sources stay moderate. Once the sample collapses towards one pole, the representation side sets the ceiling and the measurement-side tool can no longer compensate.

The limit is the data source rather than the model. The 2020-to-2024 cycles straddle a change in the platform itself, from Twitter to X, whose retuned feed amplifies divisive and partisan content. The raw 2024 signal is consistent with that amplification and with coordinated participation, though our pipeline measures neither and we do not claim them as established. The prediction-market benchmark confirms the reading from the outside: it trails the SCWG while the platform signal is usable and overtakes it once the platform's audience decouples from the electorate. A correction built on a platform's own polls inherits the platform's blind spots.

This points to the clearest line of future work. Because our stratum predictors are anchored to the platform's own distribution, the correction is scoped to within-platform skew. Anchoring the content of each stratum to an external, off-platform signal would break that circularity and let the method reach a stratum that the sample has lost, the principled fix for the cycle that resists correction here. The pipeline is ready for that extension: it is election-agnostic by design, and a new contest or platform is a registry entry rather than new code. The contribution that survives the negative result in the hardest cycle is therefore both methodological and cautionary: a reusable way to fold framing into the correction, and a usable bound on how far that correction can travel once a platform stops resembling the electorate it is meant to measure.

Appendix A

Core Implementation of the SCWG

```

1 def fit_glm(df: pd.DataFrame):
2     # 1. Define model formula mapping structural bias markers
3     formula = (
4         "positive_share ~ audience_mean_partisanship +
5         author_partisanship"
6         " + candidate_order + formality_bias + positive_ideology_score"
7         " + negative_ideology_score"
8         " + candidate_A_support_score + candidate_B_support_score"
9         " + candidate_A_oppose_score + candidate_B_oppose_score"
10        " + undirected_sentiment + sentiment_intensity + toxicity_score"
11        " + bias_confirmation + bias_anchoring + bias_availability"
12        " + bias_social_desirability + bias_acquiescence +
13        bias_demand_characteristics"
14        " + log_duration"
15        " + author_gender_male"
16        " + author_age_30_39 + author_age_40_over"
17    )
18
19    # 2. Extract total votes to serve as binomial variance weights
20    raw_weights = df["total_votes"].values.astype(float)
21
22    # 3. Fit the weighted Generalized Linear Model (Binomial family)
23
24    model = smf.glm(
25        formula=formula,
26        data=df,
27        family=sm.families.Binomial(),
28        var_weights=raw_weights,

```

```

27     )
28     result = model.fit(scale="X2")
29     return result
30
31 def build_poststrat_frame(
32     df: pd.DataFrame,
33     election: str = DEFAULT_ELECTION,
34     reference_df: pd.DataFrame | None = None,
35 ) -> pd.DataFrame:
36     ecfg = get_election_config(election)
37     scwg_cfg = ecfg["scwg"]
38
39     partisan_strata = scwg_cfg["partisan_strata"]
40     ideological_strata = scwg_cfg["ideological_strata"]
41     partisan_profiles_cfg = scwg_cfg["partisan_profiles"]
42     ideology_offsets = scwg_cfg["ideology_offsets"]
43     demographic_profiles = scwg_cfg["demographic_profiles"]
44
45     qref = reference_df if reference_df is not None else df
46
47     def _mean_shift(col: str) -> float:
48         if reference_df is None:
49             return 0.0
50         w_mean = df[col].mean()
51         g_mean = qref[col].mean()
52         if pd.isna(w_mean) or pd.isna(g_mean):
53             return 0.0
54         return float(w_mean - g_mean)
55
56     aud_shift = _mean_shift("audience_mean_partisanship")
57     auth_shift = _mean_shift("author_partisanship")
58     cons_shift = _mean_shift("positive_ideology_score")
59     neg_shift = _mean_shift("negative_ideology_score")
60
61     partisan_profiles = {}
62     for p_name, quantiles in partisan_profiles_cfg.items():
63         partisan_profiles[p_name] = {
64             "aud": qref["audience_mean_partisanship"].quantile(
65                 quantiles["aud_quantile"]) + aud_shift,
66             "auth": qref["author_partisanship"].quantile(quantiles["
67             auth_quantile"]) + auth_shift,
68             "cons": qref["positive_ideology_score"].quantile(quantiles["
69             cons_quantile"]) + cons_shift,
70             "neg": qref["negative_ideology_score"].quantile(1.0 -
71                 quantiles["cons_quantile"]) + neg_shift,
72         }

```

```

70     # Zero-Bias Counterfactual: To isolate true population preference,
    we must project a perfectly neutral polling environment (0 toxicity
    , 0 bias, neutral order)
71     median_order = 0.0
72     median_formality = 0.0
73     median_sentiment = 0.0
74     median_intensity = 0.0
75     median_toxicity = 0.0
76
77     if "log_duration" in qref.columns:
78         median_log_duration = float(qref["log_duration"].median())
79     else:
80         median_log_duration = float(np.log1p(pd.to_numeric(qref.get("
duration_minutes", pd.Series()), errors="coerce").median()))) if "
duration_minutes" in qref.columns else 0.0
81
82     bias_columns = [
83         "bias_confirmation", "bias_anchoring", "bias_availability",
84         "bias_social_desirability", "bias_acquiescence", "
bias_demand_characteristics",
85         "candidate_A_support_score", "candidate_B_support_score",
86         "candidate_A_oppose_score", "candidate_B_oppose_score",
87     ]
88     bias_medians = {b: 0.0 for b in bias_columns}
89
90     rows = []
91     for p_name, p_weight in partisan_strata.items():
92         for i_name, i_weight in ideological_strata.items():
93             profile = partisan_profiles[p_name]
94             demo = demographic_profiles[p_name]
95             mult = ideology_offsets[i_name]["cons_mult"]
96             # Mirror mult for liberal ideology: Conservative strata get
lower liberal NLI,
97             # Liberal strata get higher liberal NLI (2.0 - mult inverts
around 1.0)
98             neg_mult = 2.0 - mult
99             rows.append({
100                 "partisan_stratum": p_name,
101                 "ideology_stratum": i_name,
102                 "audience_mean_partisanship": profile["aud"],
103                 "author_partisanship": profile["auth"],
104                 "candidate_order": median_order,
105                 "formality_bias": median_formality,
106                 "undirected_sentiment": median_sentiment,
107                 "sentiment_intensity": median_intensity,
108                 "toxicity_score": median_toxicity,
109                 **bias_medians,

```

```

110         "positive_ideology_score": float(np.clip(profile["cons"]
111 ] * mult, 0.0, 1.0)),
112         "negative_ideology_score": float(np.clip(profile["neg"]
113 * neg_mult, 0.0, 1.0)),
114         "log_duration": median_log_duration,
115         "author_gender_male": demo["gender_male"],
116         "author_age_30_39": demo["age_30_39"],
117         "author_age_40_over": demo["age_40_over"],
118         "weight": p_weight * i_weight,
119     })
120
121     ps_frame = pd.DataFrame(rows)
122     logger.info(
123         f"Post-stratification frame: {len(ps_frame)} strata, "
124         f"total weight = {ps_frame['weight'].sum():.4f}"
125     )
126     return ps_frame
127
128 def poststratify(glm_result, ps_frame: pd.DataFrame) -> float:
129     predictions = glm_result.predict(ps_frame)
130     return np.average(predictions, weights=ps_frame["weight"])

```

Listing A.1. Core implementation of the SCWG: GLM fitting, post-stratification frame construction, and final aggregation.

Appendix B

LLM Prompt for cognitive bias detection

You are an expert cognitive scientist, psychometrician, and survey methodology analyst specializing in cognitive bias detection within public discourse and crowdsourced polls.

Your task is to analyze a batch of social media poll data provided in a JSON array. For each poll object, evaluate both the text prompt (how the question is framed) and the poll options (how responses are structured) to detect the presence of any of the following 6 cognitive biases:

1. Confirmation Bias: The poll text or structure seeks out, interprets, or primes information in a way that validates existing opinions/beliefs, often disregarding contradictory evidence or alternatives.
2. Anchoring: The poll text heavily relies on or introduces an initial piece of information (the anchor), disproportionately influencing how a respondent might value subsequent choices.
3. Availability Heuristic: The question prompts judgments based on immediate, sensational, or highly memorable events/information currently in public memory rather than objective data.
4. Social Desirability Bias: The poll options or phrasing pressure respondents to choose options that will be viewed favorably by others, over-reporting good behavior or under-reporting undesirable behavior.

```

5. Acquiescence Bias: The question or statement induces yea-
   saying, nudging the respondent to passively agree with a
   premise (e.g. Do you agree that...?) regardless of their
   true view.
6. Demand Characteristics Bias: The poll contains clear clues,
   leading words, or surveyor signals indicating what the
   surveyor wants to hear, causing respondents to adjust
   answers accordingly.

### Instructions:
- Process every item in the provided JSON array.
- For each item, return its original "id".
- Identify which biases are present. If no biases are found,
  return an empty array for 'biases_detected'.
- Do not add conversational text, introductory remarks, or
  markdown wrapping outside of the valid JSON object. Your
  output must strictly be a parseable JSON array.

### JSON Output Schema:
[
  {
    "id": <int/string matching input id>,
    "biases_detected": [
      {
        "bias_type": "Name of Bias (or 'None')",
        "confidence_score": <float between 0.0 and 1.0>,
        "evidence": "Quote string from the text or options
showcasing this bias",
        "rationale": "Brief scientific explanation of why this
bias applies here"
      }
    ]
  }
]

```

Listing B.1. System instruction prompt provided to the LLM for cognitive bias detection.

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